Supporting Information

Conformational landscapes in the ground and excited states dictate circularly polarized excimer emission: A thermodynamic–kinetic interplay in carbazole dimer systems

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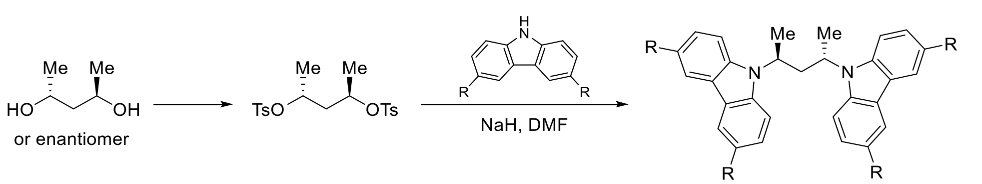
Experimental Details

All reactions were carried out in dry solvents under a nitrogen atmosphere; medium-pressure column chromatography was performed on silica gel (Shoko Science, Purif-Pack-EX SI-25) using a Yamazen system equipped with a PREP UV-10V detector and a Model 540 pump; HPLC analysis was conducted on a JASCO LC-2000 Plus series system comprising a quaternary pump (PU-2089 Plus), column thermostat (JASCO CO-2067 Plus), circular dichroism detector (JASCO CD-4095), and UV detector (JASCO UV-2075 Plus) at a column temperature of 20 °C with peak detection at 340 nm; ^{1}H NMR (600 MHz) and ^{13}C{^{1}H} NMR (150 MHz) spectra were recorded on a Bruker Avance spectrometer; electronic absorption (UV–Vis), circular dichroism (CD), fluorescence (FL), and circularly polarized luminescence (CPL) spectra were measured in non-degassed methylcyclohexane solutions (1×10^{-5} M for UV–Vis, CD, and FL; 1×10^{-4} M for CPL) using a standard 1 cm quartz cuvette fitted with a temperature controller, where UV–Vis spectra were obtained on a JASCO V-660 spectrophotometer (bandwidth = 1 nm, scan rate = 40 nm min^{-1}, response = medium, data interval = 0.1 nm), CD spectra on a JASCO J-820YH spectropolarimeter (bandwidth = 1 nm, scan rate = 50 nm min^{-1}, response time = 8 s, 4 accumulations, data interval = 0.1 nm), FL spectra on a JASCO FP-8500 spectrofluorometer (excitation wavelength = 300 nm, excitation and emission slit widths = 1 nm, scan rate = 50 nm min^{-1}, response = medium, data interval = 0.1 nm), and CPL spectra on a JASCO CPL-300 spectropolarimeter (excitation and emission slit widths = 16 nm, scan rate = 50 nm min^{-1}, response time = 4 s, 2 accumulations, data interval = 0.1 nm). Fluorescence lifetimes were measured in aerated methylcyclohexane solution using a time-correlated single-photon counting (TCSPC) system (Hamamatsu Photonics, C16361-01) with a 1 cm quartz cuvette. Emission was collected at the long-wavelength edge of the excimer emission band (420 or 430 nm) following excitation with a pulsed LED (λ_{ex} = 280 nm). Decay profiles were deconvoluted by iterative reconvolution using a monoexponential fitting model, yielding lifetimes with a goodness-of-fit χ^2 value between 0.9 and 1.1.

Preparation of (*R*,*R*)- and (*S*,*S*)-2,4-di-*N*-carbazolylpentane (1a) and the *tert*-butyl derivative (1b)



1a: R = H
b: R = *t*-Bu

In a dry flask, (*S,S*)- or (*R,R*)-2,4-pentanediol (0.51 g, 4.9 mmol) and pyridine (3 mL) were combined. The solution was cooled to 0 °C, and a pyridine solution (3 mL) of *p*-toluenesulfonyl chloride (1.94 g, 10.0 mmol, 2.0 equiv) was added dropwise. The reaction mixture was stirred at 0 °C and then allowed to warm to room temperature, where it was stirred for 40 h. Upon completion (monitored by TLC), the mixture was quenched with aqueous hydrochloric acid (1 M) and extracted with dichloromethane (3 × 10 mL). The combined organic layers were dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced pressure. The crude product was recrystallized from a mixture of hexane and ethyl acetate (9 : 1 v/v), and the resulting white solid, (*S,S*)- or (*R,R*)-2,4-bis(tosyloxy)pentane, was collected by vacuum filtration (1.51 g, 75% yield). This material was used directly in the next step without further purification.

In a separate dry flask under a nitrogen atmosphere, 9*H*-carbazole (0.37 g, 2.2 mmol, 1.1 equiv) and sodium hydride (55% dispersion in paraffin oil, 0.21 g, 8.8 mmol, 4.4 equiv) were suspended in anhydrous DMF (4 mL). The mixture was stirred at 60 °C for 1 h to generate the carbazolyl anion, then cooled to ambient temperature. A DMF solution of 2,4-bis(tosyloxy)pentane (0.41 g, 1.0 mmol) was added dropwise, and the resulting mixture was heated again to 60 °C and stirred for an additional 1 h. After completion (TLC monitoring), the reaction was quenched with saturated aqueous ammonium chloride solution and extracted with a mixture of hexane and ethyl acetate (3 × 10 mL). The combined organic layers were dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced pressure. The crude product was purified by flash column chromatography on silica gel using hexane/ethyl acetate (99 : 1 v/v) as eluent to afford **1a** as a white solid (0.043 g, 11% yield). A similar procedure was employed for the preparation of the *tert*-butyl derivative, using 3,6-di-*tert*-butyl-9*H*-carbazole (0.62 g, 2.2 mmol, 1.1 equiv) in place of 9*H*-carbazole, to afford **1b** as a white solid (0.043 g, 7% yield).

Computational Details

Ground state geometries were optimized using density functional theory (DFT) with ωB97X-D[39] functional and the def2-TZVP[40] basis set, as implemented in the Gaussian 16 (Revision C.01)[41] suite of program with default thresholds. Harmonic vibrational frequency analyses at the same level of theory confirmed that the optimized geometries are energy minima, as indicated by the absence of imaginary frequencies. For the excited state, the geometry optimizations were carried out at PBE0/def2-TZVP level of theory,[42] as PBE0 functional generally provides improved performance for excited state calculations.[43] The optimized ground- and excited-state geometries are given in **Tables S4** and **S5**, respectively. Potential energy surfaces (PES) for ground and excited states were computed at the ωB97X-D/def2-SVP level by scanning the two dihedral angles ϕ_1 (N1-C2-C3-C4) and ϕ_2 (N2-C4-C3-C2) along the tethering –N(Cbz)–C–CH_2–C– N(Cbz)– linkage. Both angles were varied from –180° to +180° in 10° increments, generating a two-dimensional grid of 1296 single-point energy calculations for each electronic state. For the *tt* conformer of **1b**, geometry optimization using default convergence parameters converged to the global minimum (#1), with dihedral angles (ϕ_1 = 33°, ϕ_2 = 33°) that differ significantly from those observed in the crystal structure (ϕ_1 = 52°, ϕ_2 = 58°). By reducing the optimization step size, a local minimum (#2) was also located, featuring a geometry closely resembling the single-crystal structure (ϕ_1 = 47°, ϕ_2 = 59°). Energy decomposition analyses (EDA) were performed using symmetry-adapted perturbation theory (SAPT)[44] at the SAPT0 level with the jun-cc-pVDZ[45] basis set, as implemented Psi4.[46] The intramolecular interaction between the two carbazole moieties was evaluated using the F/I-SAPT0 method. The total interaction energy (E_{tot}) is decomposed into four components:

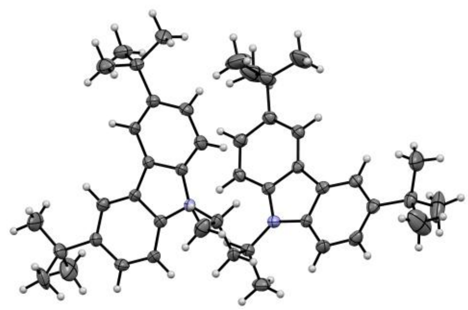
$$E_{tot} = E_{ele} + E_{ex} + E_{ind} + E_{dis}$$

where E_{ele}, E_{ex}, E_{ind}, and E_{dis} represent the electrostatic, exchange (Pauli repulsion), induction, and dispersion contributions, respectively. Non-covalent interaction (NCI) analysis[47] was employed to visualize the weak interactions between the carbazole fragments.

X-Ray Crystallographic Analysis

Colorless plate-shaped single crystals of **1b** were grown by slow diffusion of ethanol into a solution of **1b** in toluene. Intensity data were collected at 153 K on a Rigaku XtaLAB Synergy-S/Mo diffractometer (MoK_α radiation: λ = 0.71073 Å) equipped with PhotonJet (Mo) source and HyPix-6000HE photon counting detector. The structure was solved by intrinsic phasing method (SHELXT 2018/2)[48] and refined by full-matrix least-squares on F^2 (SHELXL 2098/3).[49] All hydrogen atoms were placed using AFIX instructions, while all non-hydrogen atoms were refined anisotropically. The crystal data are summarized in **Table S1**. CCDC 2536729 contains the supplementary crystallographic data for compound **1b**. This data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) at https://www.ccdc.cam.ac.uk/structures/.

Table S1. Single Crystal Structure in Thermal Ellipsoid Plot (50% Probability for Thermal Ellipsoids; Gray: Carbon; Blue: Nitrogen; White: Hydrogen) and Crystallographic Data of **1b**.



Empirical Formula	$C_{45}H_{58}N_2$	Volume / Å^3	1909.04(13)
Formula Weight	626.93	Z	2
Crystal Description	Plate	D_x / g cm^{-3}	1.091
Crystal color	Colorless	μ / mm^{-1}	0.062
Crystal Size / mm	0.30 × 0.15 × 0.01	2ϑ / °	61.688
Temperature / K	153	Reflections Collected	33761
Crystal System	$P2_1$ (#4)	Independent Reflections	9605
Space Group	P21	R_{int}	0.0647
a / Å	12.4887(5)	R_1 (I > 2σ(I))	0.0734
b / Å	9.6783(3)	R_1 (all data)	0.1156
c / Å	16.1679(7)	wR_2 (I > 2σ(I))	0.1763
α / °	90	wR_2 (all data)	0.1933
β / °	102.342(4)	GOF (all data)	1.045
γ / °	90		

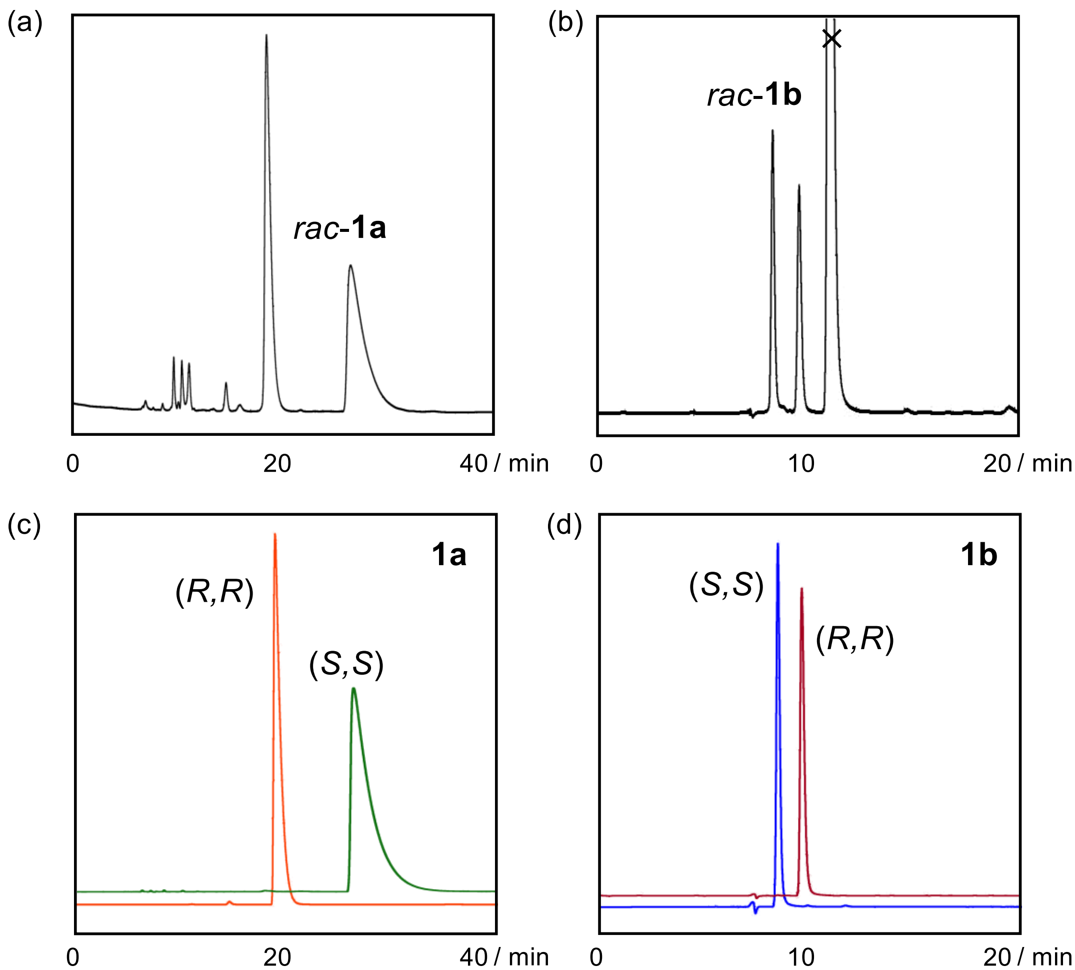


Figure S1. HPLC profiles of (a,b) racemic and (c,d) optically pure carbazole dimers **1a** and **1b**. Chiral stationary phase: DAICEL CHIRALPAK® IB N-5; eluent: hexane/ethanol = 95 : 5 for **1a**, hexane/toluene = 95 : 5 for **1b**; flow rate: 0.5 mL min^{-1}.

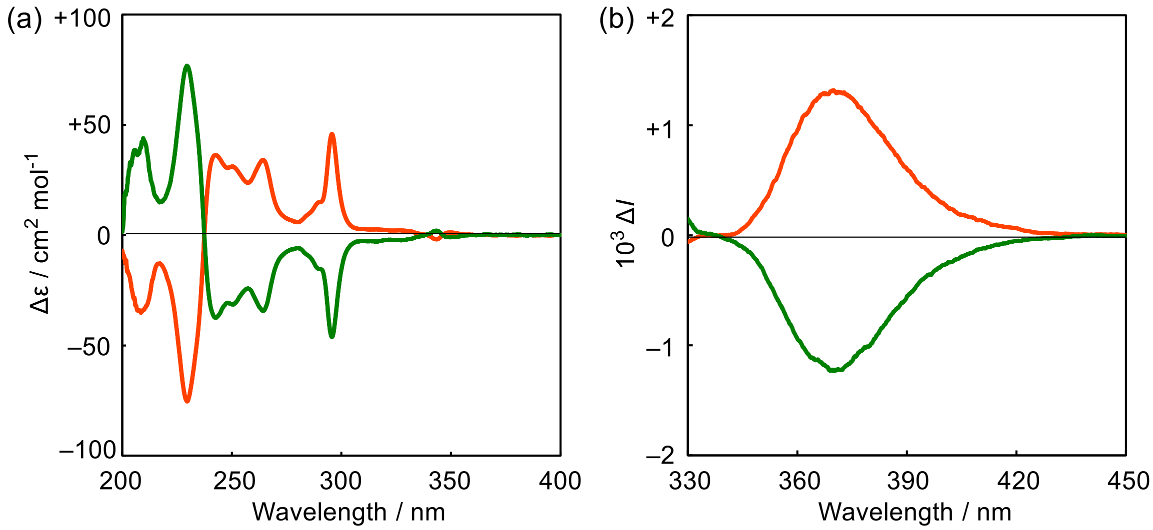


Figure S2. (a) CD and (b) CPL spectra of (*S*,*S*)-**1a** (green) and (*R*,*R*)-**1a** (orange) in methylcyclohexane at 25 °C.

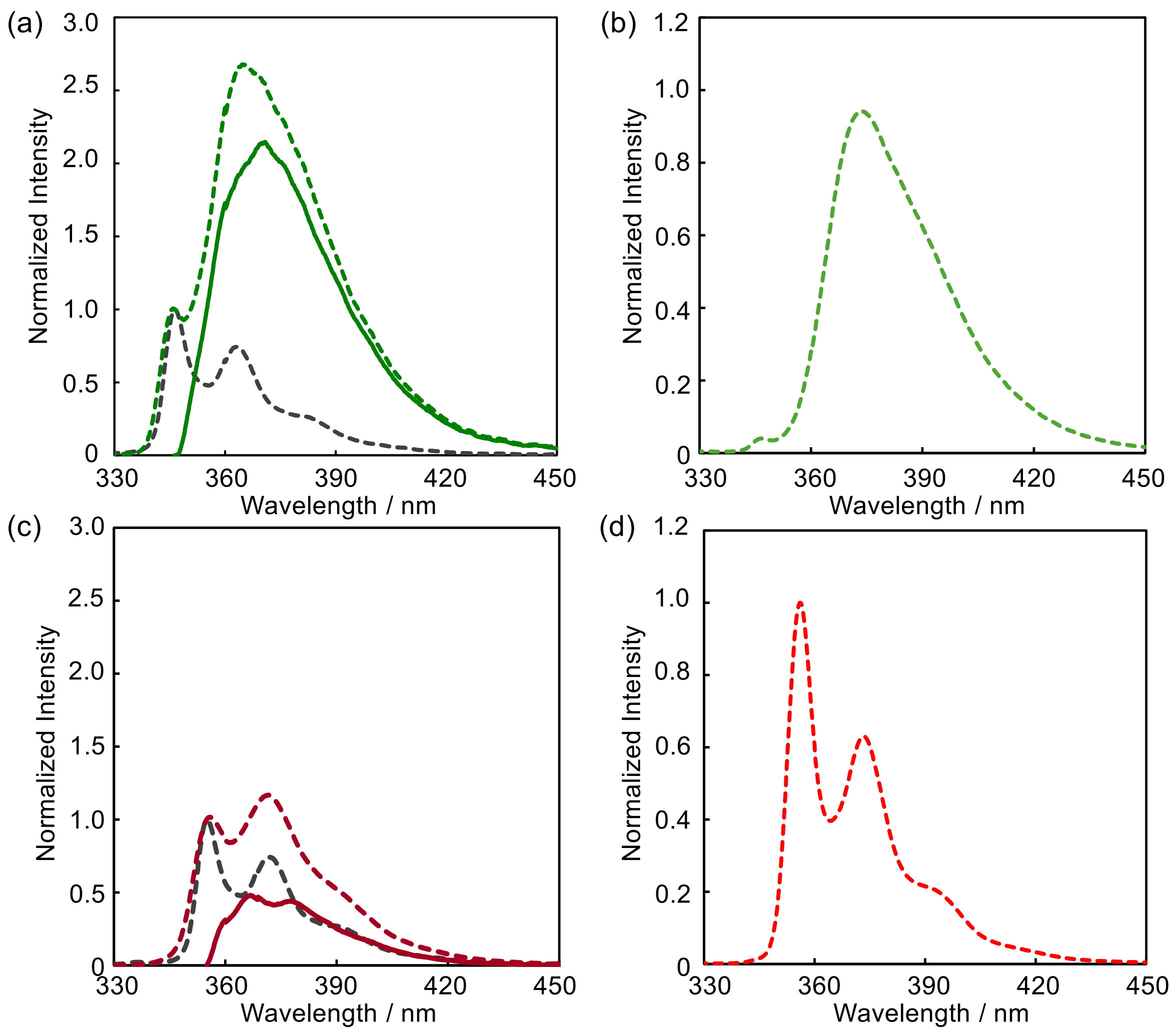


Figure S3. Steady-state fluorescence spectra of carbazole dimers (a,b) **1a** and (c,d) **1b** in methylcyclohexane at +25 (left) and −110 °C (right). The excimer emission profiles at 25 °C (solid lines) were obtained by subtracting the monomer fluorescence component (black dotted lines) from the total observed emission.

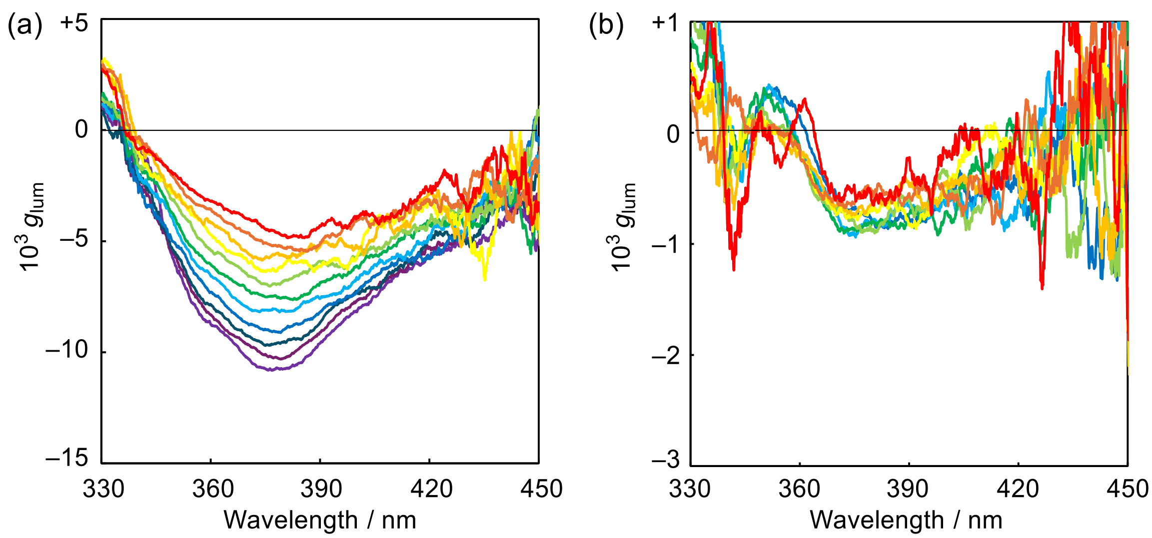


Figure S4. Temperature dependence of the g_{lum} profiles of carbazole dimers (a) **1a** (−110 to +90 °C) and (b) **1b** (−50 to +70 °C) in methylcyclohexane.

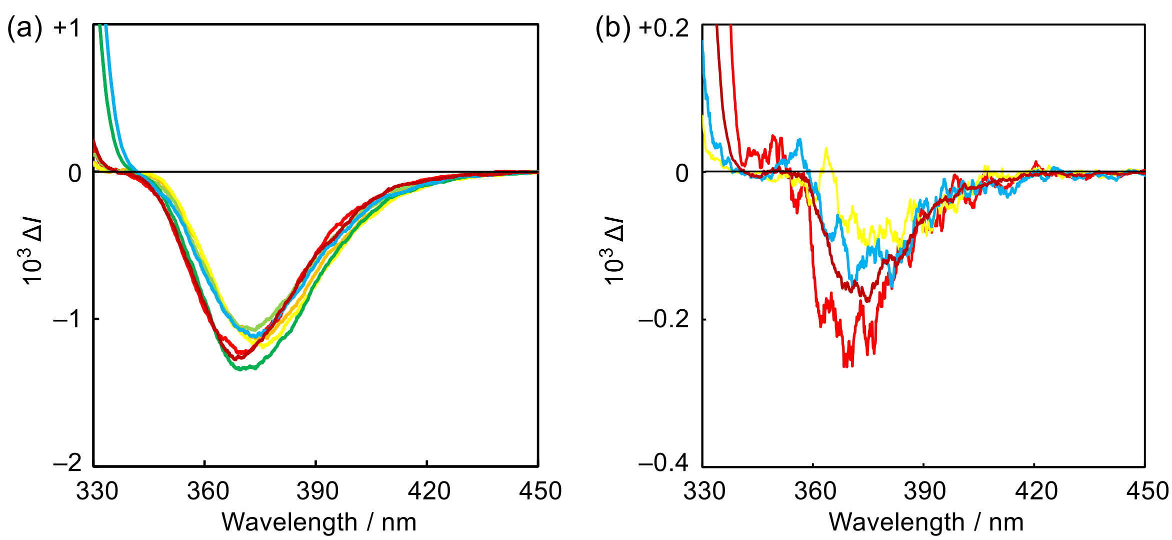


Figure S5. Solvent dependence of the CPL spectra of carbazole dimer **1a** (left) and **1b** (right) at 25 °C. Colors correspond to the following solvents: brick red – methylcyclohexane; red – hexane; orange – toluene; yellow – dichloromethane; light green – tetrahydrofuran; green – methanol; light blue – acetonitrile.

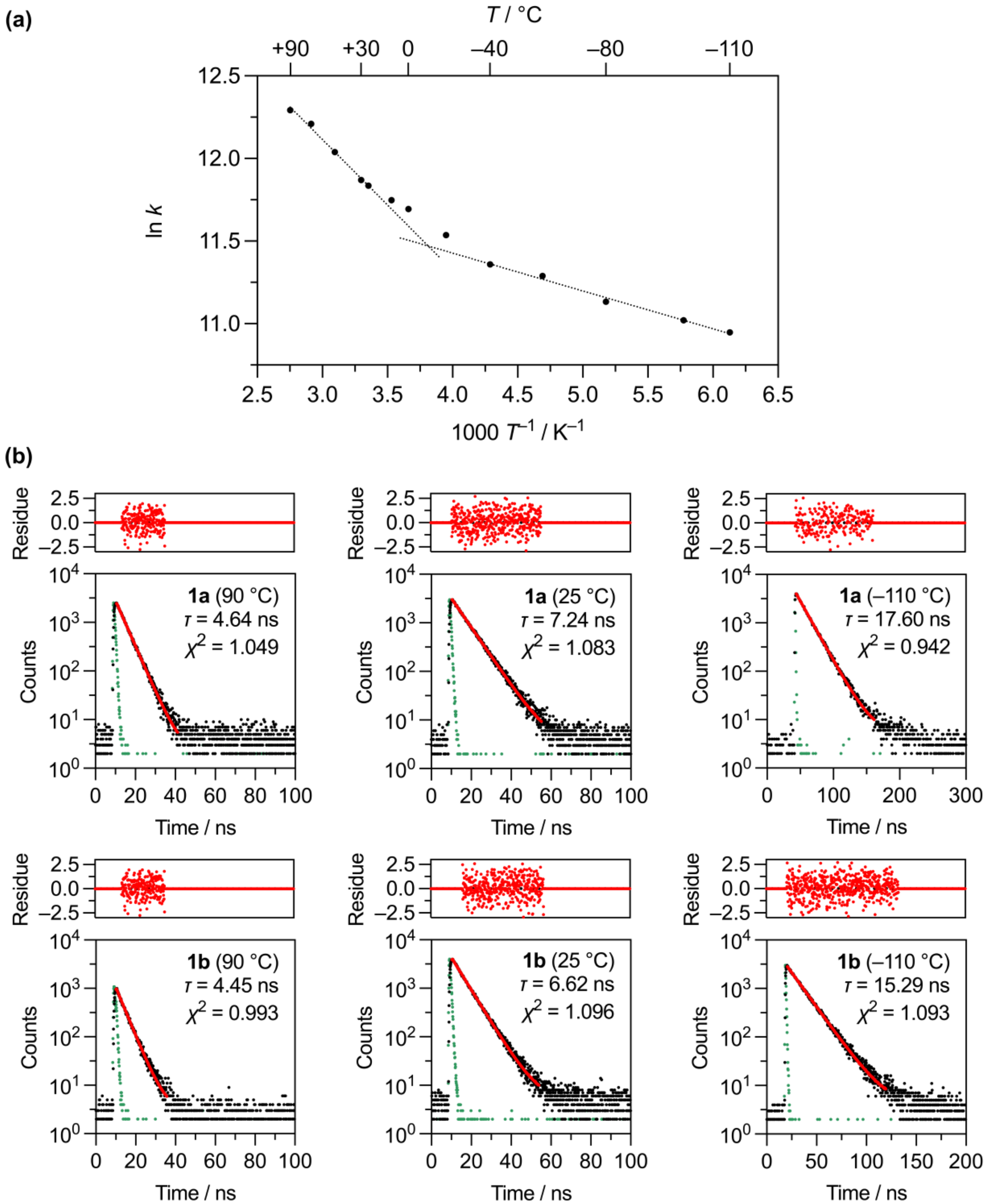


Figure S6. Temperature-dependent fluorescence kinetics of carbazole dimers **1a** and **1b** in methylcyclohexane. Top: Fluorescence decay rate ($k = 1/\tau$) for **1a** plotted against reciprocal temperature ($1/T$). Eyring-type analysis of the two apparent linear segments yields differential apparent activation enthalpies ($\Delta\Delta H^{\ddagger}_{app}$) of 6.6 kJ mol^{-1} (high-temperature region) and 1.9 kJ mol^{-1} (low-temperature region). Middle and bottom: Representative fluorescence decay curves for **1a** (middle) and **1b** (bottom) at –110, 25, and 90 °C. Black traces: experimental decays; green traces: instrument response function (IRF); red traces: monoexponential fits.

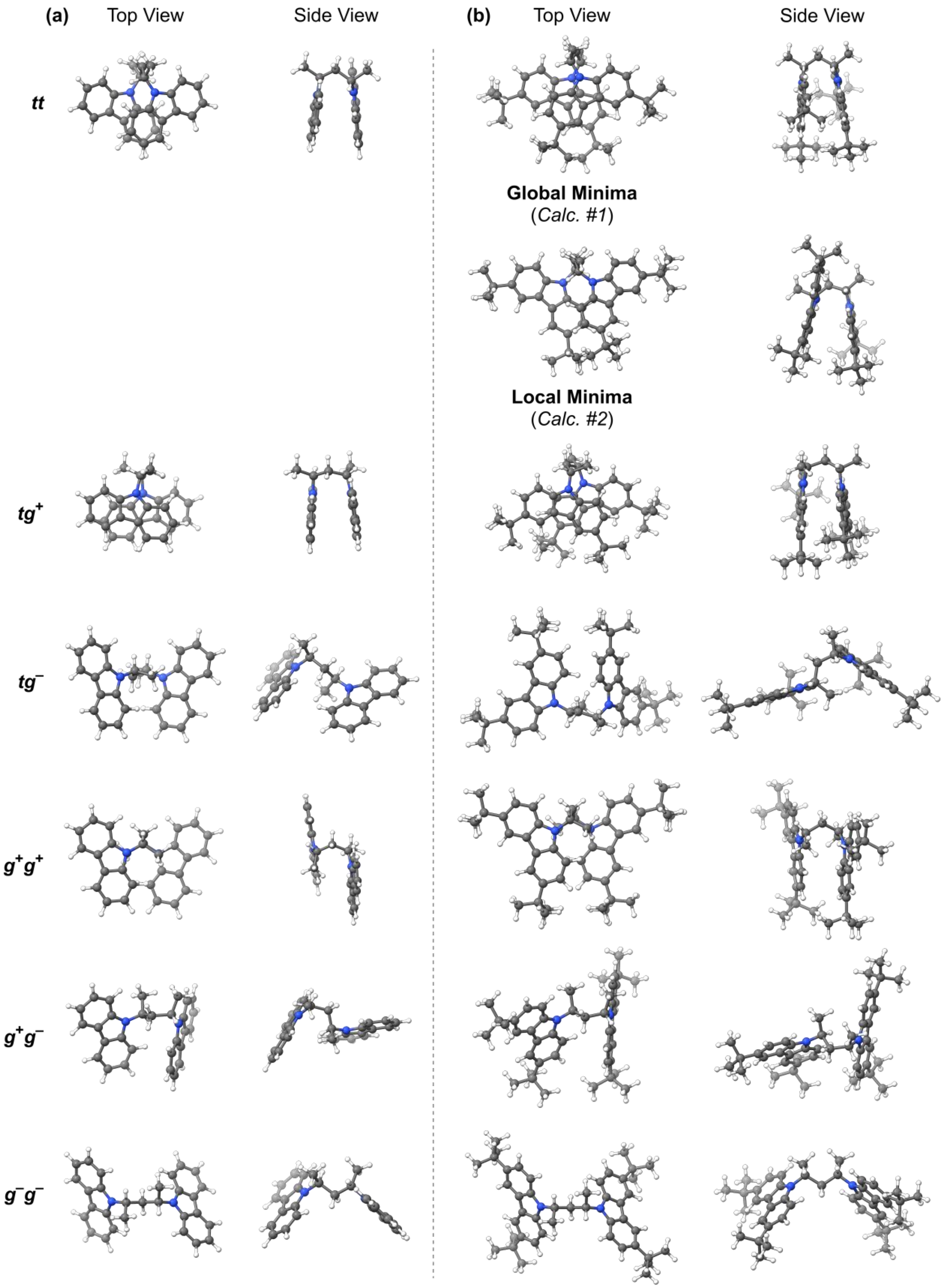


Figure S7. Calculated structures of all possible conformers of carbazole dimer (a) **1a** and (b) **1b** in the ground (S_0) state.

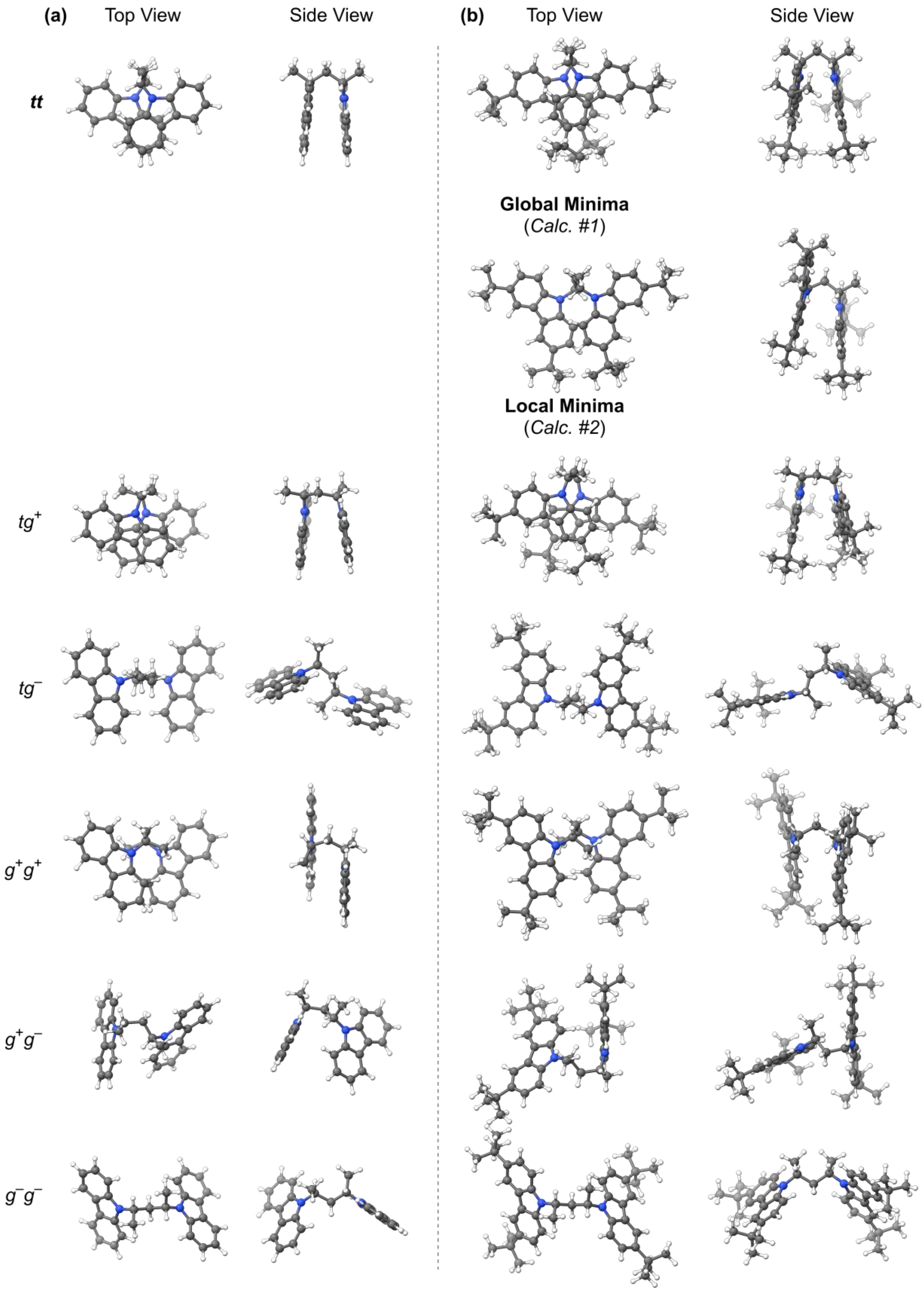


Figure S8. Calculated structures of all possible conformers of carbazole dimer (a) **1a** and (b) **1b** in the ground (S_1) state.

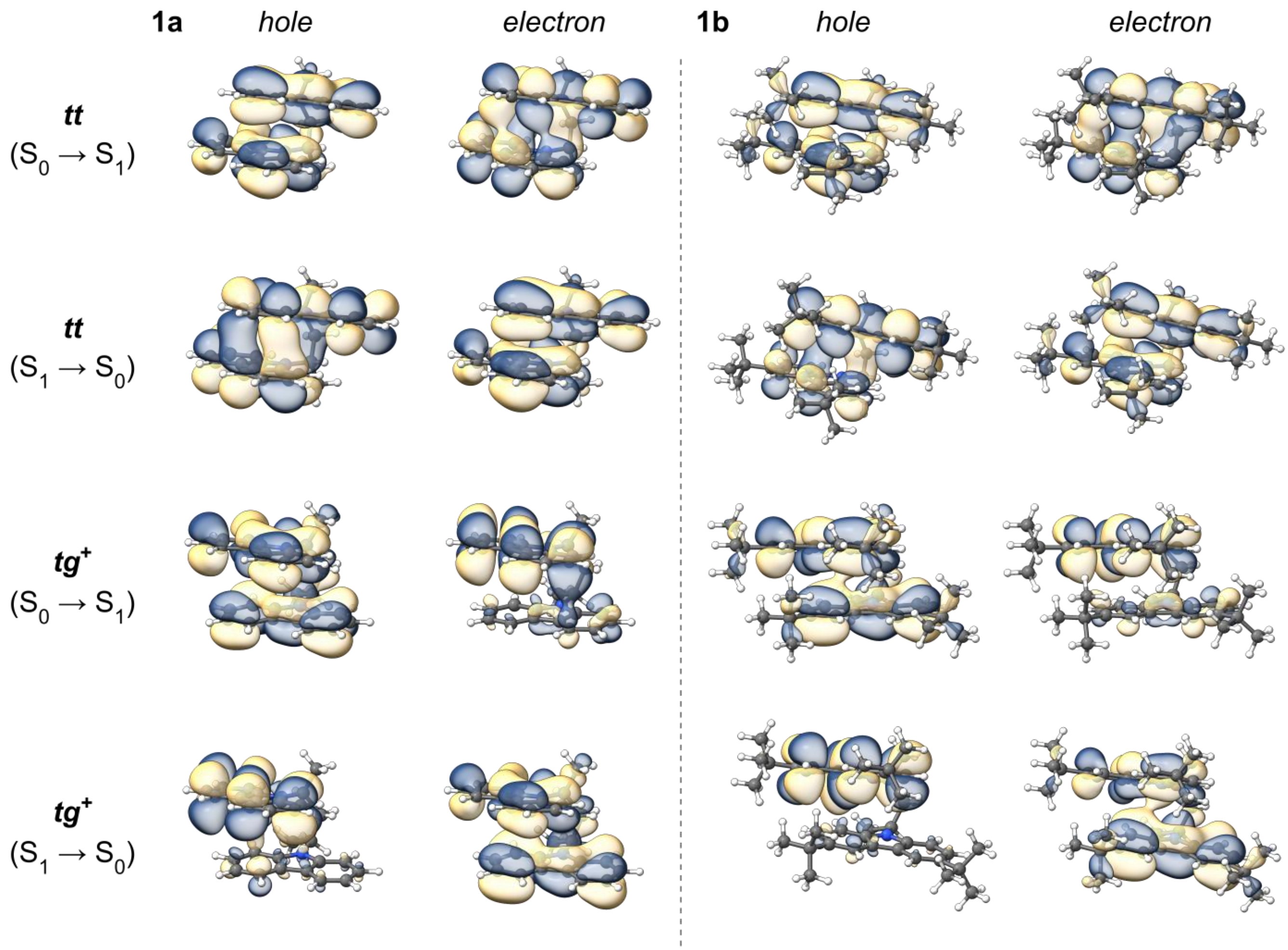


Figure S9. Natural transition orbital (NTO) analysis for the S_0 to S_1 excitation (top) and S_1 to S_0 emission (bottom) of the *tt* and *tg*$^+$ conformers of carbazole dimers **1a** (left) and **1b** (right). Isovalue = 0.015.

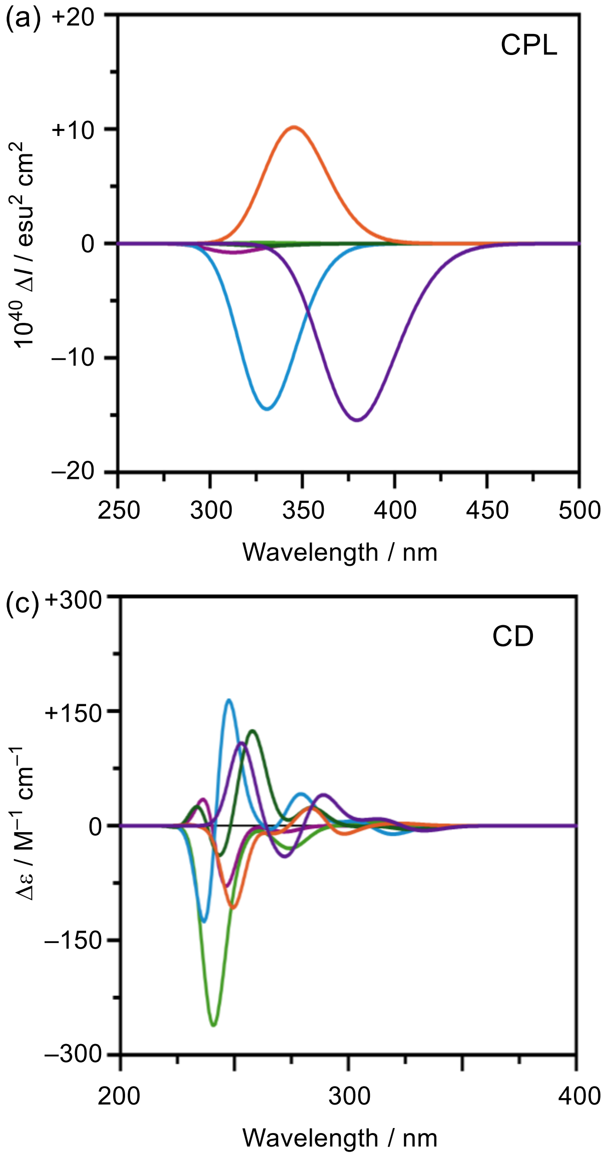


Figure S10. Calculated CPL (top) and CD (bottom) spectra of individual conformers of carbazole dimers (a,c) **1a** and (b,d) **1b.** Color code: purple – *tt*; orange – tg^+; green – tg^-; light blue – g^+g^+; light green – g^+g^-; light purple – g^-g^-. The spectra were obtained using calculated rotational strengths in the length gauge, broadened with Gaussian functions with a full width at 1/e height fixed at 0.5 eV.

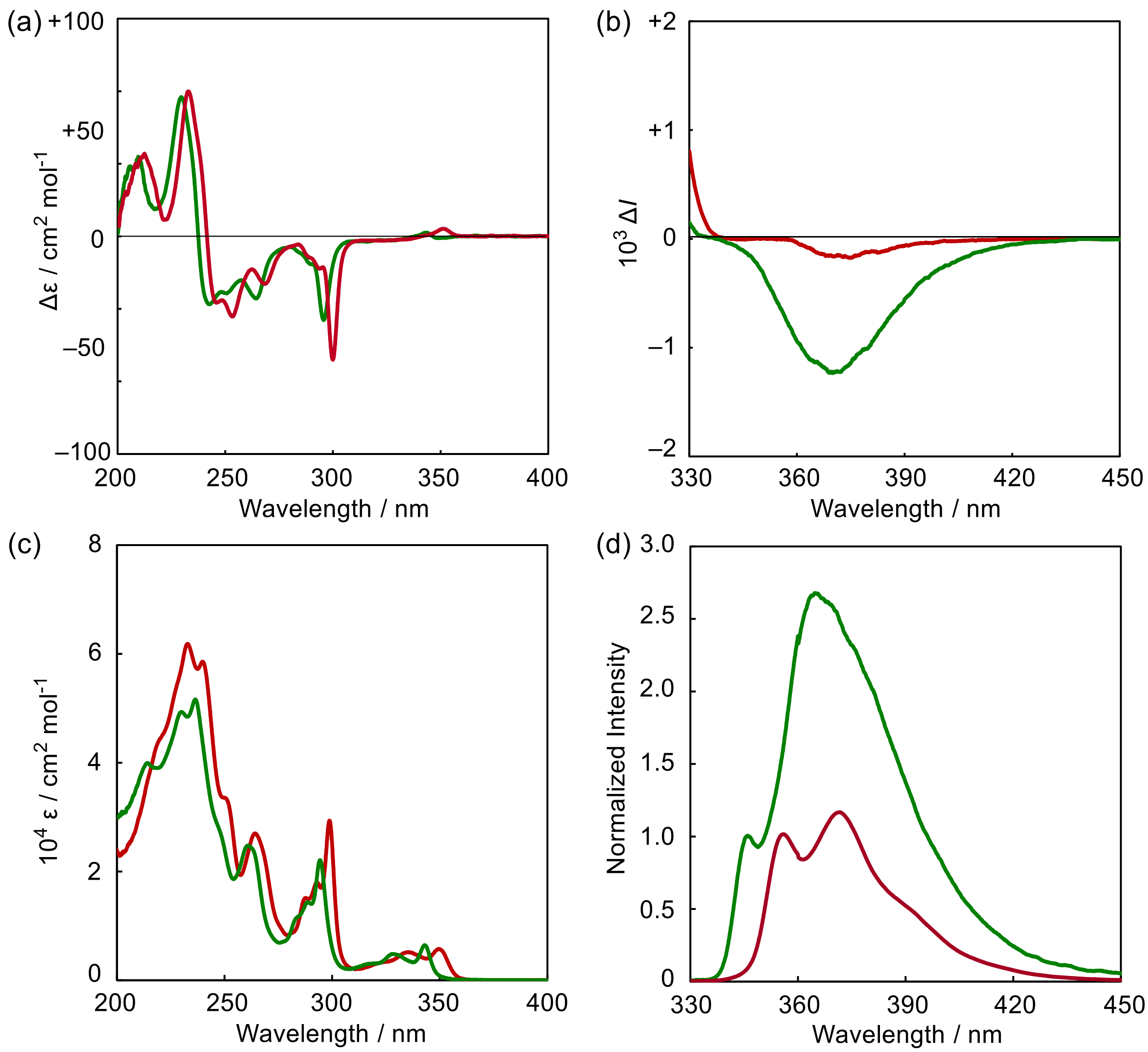


Figure S11. Comparison of (a) CD and (b) CPL spectra of (*S,S*)-enantiomers of carbazole dimer **1a** (green) and **1b** (red) in methylcyclohexane at 25 °C. Comparison of (c) UV-Vis absorption and (d) fluorescence spectra of **1a** (green) and **1b** (red) under identical conditions.

Table S2. Detailed Computed Structural and Spectral Parameters for Individual Conformers of Carbazole Dimers **1a** and **1b** in the S_0 and S_1 States [a]

State	Conf.	ΔE	ϕ_1 / °	ϕ_2 / °	ψ / °	d / Å	E_{01} / eV (nm)	R	D	ϑ / °	$10^3\ g$
1a											
S_0	tt	≡ 0.0	37.2	37.2	–121	4.09	3.73 (333)	–32.3	15500	180	–8.3
	tg^+	+15.4	56.3	–75.9	+153	3.76	3.90 (318)	+15.5	12700	68	+4.9
	tg^-	+46.9	68.1	176.4	+33	7.11	4.01 (310)	–9.6	31300	98	–1.2
	g^+g^+	+78.3	–67.3	–67.3	+48	6.62	4.09 (303)	–31.7	42100	135	–3.0
	g^+g^-	+64.2	158.9	–40.7	–70	6.05	4.08 (304)	+13.6	13900	74	+3.9
	g^-g^-	+52.7	175.2	175.2	+126	7.58	4.04 (307)	–0.7	33200	91	–0.1
S_1	tt	≡ 0.0	39.5	39.5	–112	4.19	3.27 (380)	–44.9	5370	180	–33
	tg^+	+34.5	50.0	–85.7	+135	4.24	3.59 (346)	+29.9	14600	57	+8.2
	tg^-	+20.9	69.1	169.7	+25	7.28	3.76 (330)	–0.6	27000	92	–0.1
	g^+g^+	+57.2	–58.3	–58.3	+65	6.15	3.75 (331)	–39.3	40700	138	–3.9
	g^+g^-	+35.1	138.1	–62.9	–12	5.82	3.86 (322)	–1.9	29600	92	–0.3
	g^-g^-	+23.0	173.7	174.5	+126	7.57	3.77 (328)	+0.1	28700	90	+0.02
1b											
S_0	tt	≡ 0.0	33.2	33.0	–129	3.74	3.66 (338)	–18.4	19900	160	–3.7
	[tt	+31.1	47.0	59.2	–129	5.52	3.91 (317)	+12.5	22400	79	+2.2]
	tg^+	+7.2	53.7	–78.3	+148	3.95	3.82 (325)	+2.2	19900	86	+0.5
	tg^-	+50.6	74.7	–168.4	+55	6.61	3.82 (324)	–18.2	21000	102	–3.5
	g^+g^+	+53.3	–61.5	–61.5	+58	6.22	3.93 (316)	–45.4	32300	126	–5.6
	g^+g^-	+54.8	161.2	–41.0	–67	6.19	3.98 (311)	+18.4	19100	78	+3.9
	g^-g^-	+65.0	175.5	175.7	+126	7.59	3.99 (311)	–4.6	35100	92	–0.5
S_1	tt	≡ 0.0	33.5	36.2	–123	4.07	3.38 (366)	–32.9	15000	151	–8.8
	[tt	+4.7	61.0	66.5	–109	6.19	3.74 (331)	+11.6	38700	80	+1.2]
	tg^+	+13.5	55.3	–84.7	+141	4.30	3.56 (348)	+13.9	13900	66	+4.0
	tg^-	–0.7	72.7	168.2	+25	7.36	3.64 (340)	–0.9	26500	91	–0.1
	g^+g^+	+44.0	–70.2	–71.2	+43	6.89	3.74 (331)	–6.7	40200	97	–0.7
	g^+g^-	+26.1	150.0	–49.2	–87	5.74	3.52 (353)	+2.0	1150	71	+6.9
	g^-g^-	+13.1	173.8	174.7	+122	7.59	3.74 (332)	–0.9	29400	91	–0.1

[a] All energy calculations were performed at the ωB97X-D/def2-TZVP level of theory on geometries optimized as described in the Computational Details section. The following geometric and photophysical parameters of fully optimized (unconstrained) structures are compared: relative electronic energy (ΔE, in kJ mol^{-1}), dihedral angles ϕ_1 and ϕ_2 around the tethering –C–CH_2–C– bonds, interplane twist angle (ψ), centroid-to-centroid distance (d) between carbazole units, vertical excitation energy (E_{01}, in eV and nm), rotational strength (R, in 10^{-40} cgs units), electric dipole strength (D, in 10^{-40} cgs units), angle between transition dipole moments (ϑ), and absorption or luminescence dissymmetry factors (g_{abs} or g_{lum}). [b] Values for the second-lowest-energy local minimum structure are given in square brackets.

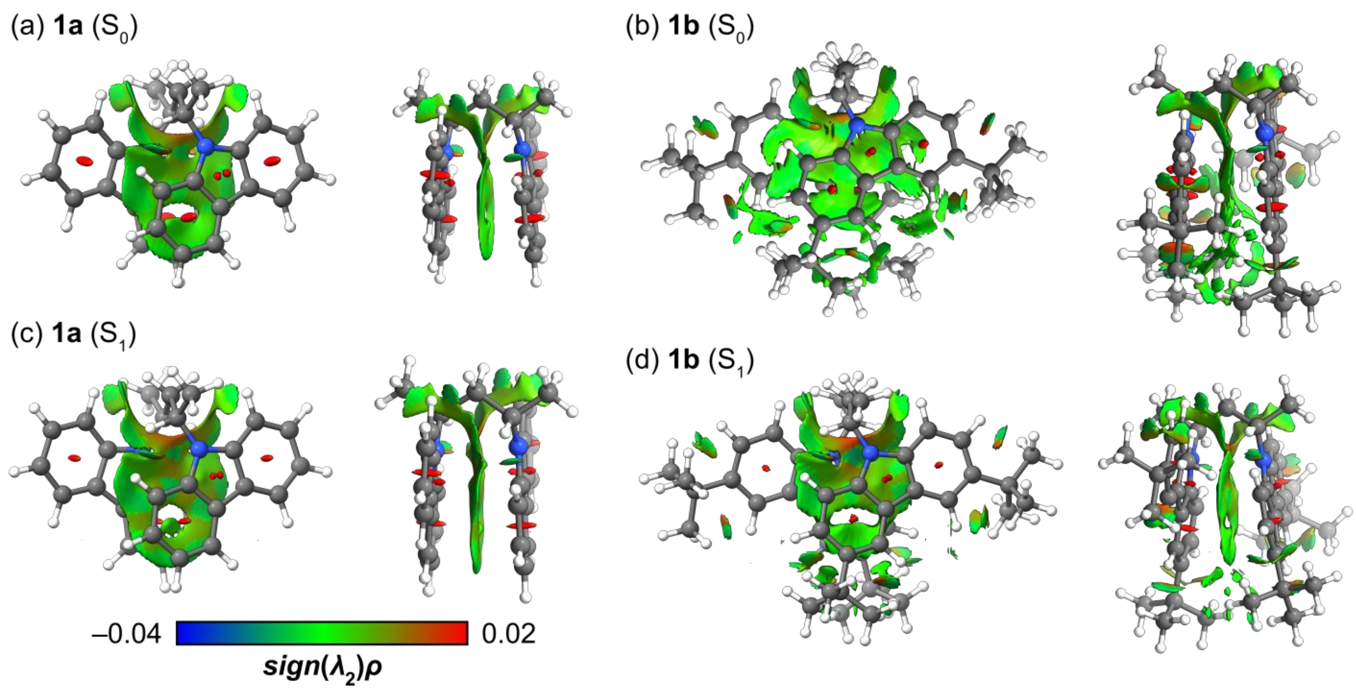


Figure S12. Non-covalent Interaction (NCI) plots (RDG = 0.05) for the *tt* conformers of carbazole dimers (a, c) **1a** and (b, d) **1b** in the S_0 and S_1 states.

Table S3. Results from energy decomposition analysis (EDA) of weak interactions between the two carbazole moieties in **1a** and **1b** in the S_0 and S_1 states. Energies are reported in kJ mol^{-1}.

Compound	State	E_{tot}	E_{ele}	E_{ex}	E_{ind}	E_{dis}
1a	S_0	–32.7	–8.8	60.6	–9.4	–75.1
	S_1	–18.0	–40.6	135.7	–12.3	–100.7
1b	S_0	–57.7	–20.8	107.2	–13.9	–130.2
	S_1	–45.0	–20.9	94.4	–13.3	–105.2

Table S4. Optimized Geometries of Carbazole Dimers **1a** and **1b** in the Ground (S_0) State at the ωB97X-D/def2-TZVP Level.

1a (tt) (-1230.336737 hartree)			
C	0.159058	-3.086799	2.044292
C	-0.790287	-2.127484	2.402764
C	-0.594668	-0.783704	2.147326
C	0.585266	-0.409876	1.512451
C	1.553601	-1.364188	1.158091
C	1.332795	-2.710452	1.421291
C	2.254613	0.717263	0.524409
C	3.087539	1.684310	-0.034426
C	4.274146	1.267171	-0.610924
C	4.641749	-0.079527	-0.632423
C	3.821743	-1.035871	-0.062615
C	2.625619	-0.642081	0.522047
C	0.232885	2.058563	1.289627
C	0.000020	2.857871	-0.000035
C	-0.232926	2.058541	-1.289676
C	-2.254611	0.717211	-0.524431
C	-0.585205	-0.409942	-1.512358
C	-3.087541	1.684266	0.034388
C	-2.625623	-0.642131	-0.522075
C	0.594755	-0.783778	-2.147177
C	-1.553578	-1.364244	-1.158069
C	-4.274174	1.267139	0.610843
C	-3.821763	-1.035910	0.062565
C	0.790359	-2.127554	-2.402643
C	-1.332788	-2.710504	-1.421299
C	-4.641783	-0.079556	0.632334
C	-0.159025	-3.086858	-2.044248
C	0.813941	2.942120	2.389269
C	-0.814088	2.942078	-2.389285
N	1.012868	0.846865	1.113564
N	-1.012843	0.846810	-1.113543
H	1.708461	-2.440203	-2.883862
H	-4.932966	2.005328	1.050847
H	-4.103929	-2.081565	0.075506
H	-2.067546	-3.452433	-1.134273
H	-5.576662	-0.371511	1.092492
H	0.031480	-4.131460	-2.253030
H	1.819292	3.278780	2.130727
H	0.870747	2.390702	3.327867
H	0.187671	3.822865	2.541967
H	-0.746577	1.714223	1.617450
H	0.746520	1.714234	-1.617581
H	1.358942	-0.066457	-2.414282
H	-2.830514	2.733901	0.022252
H	-0.845981	3.521657	0.191600
H	0.846076	3.521578	-0.191690
H	2.830535	2.733951	-0.022266
H	4.932927	2.005356	-1.050951
H	5.576608	-0.371488	-1.092614
H	4.103906	-2.081527	-0.075544
H	-0.031463	-4.131402	2.253048
H	-1.708364	-2.440127	2.884034
H	-1.358832	-0.066377	2.414479
H	2.067516	-3.452389	1.134188
H	-0.187878	3.822864	-2.541989
H	-1.819452	3.278669	-2.130708
H	-0.870888	2.390670	-3.327889

1a (tg^+) (-1230.331247 hartree)			
C	1.686355	-0.488148	2.737125
C	1.062044	-1.731573	2.853933
C	-0.209272	-1.923137	2.350861
C	-0.851499	-0.866173	1.717737
C	-0.205706	0.375827	1.580571
C	1.068299	0.574836	2.108415
C	-2.141179	-0.735130	1.091057
C	-2.212083	0.577877	0.593484
C	-3.198520	-1.612594	0.894464
C	-4.316729	-1.175949	0.209804
C	-4.383239	0.133736	-0.269879
C	-3.343071	1.026267	-0.083066
C	-0.772750	2.627288	0.537337
C	0.494486	2.848269	-0.295246
C	0.642974	2.102493	-1.634539
C	2.218204	0.327321	-0.960336
C	2.258381	-1.061063	-0.762385
C	0.984004	-1.568927	-1.193750
C	0.223596	-0.465307	-1.628506
C	3.316222	1.125380	-0.647566
C	4.435073	0.510121	-0.115108
C	4.475982	-0.868451	0.101705
C	3.390995	-1.658919	-0.227162
C	0.459823	-2.854866	-1.231639
C	-0.817658	-3.046464	-1.715302
C	-1.562204	-1.955108	-2.163521
C	-1.063518	-0.667852	-2.127243
C	-0.517582	2.372013	-2.579650
C	-0.766578	3.505354	1.787809
N	-1.029738	1.238453	0.873405
N	0.983491	0.692308	-1.471201
H	2.684911	-0.353286	3.132119
H	1.579884	-2.545759	3.343916
H	-0.699718	-2.884658	2.439267
H	1.585467	1.518527	2.024929
H	-3.140123	-2.629148	1.263140
H	-5.148580	-1.848308	0.045387
H	-5.270641	0.462896	-0.795990
H	-3.424806	2.037353	-0.460221
H	-1.621203	2.938079	-0.071128
H	0.530615	3.917612	-0.524100
H	1.372651	2.646039	0.316041
H	1.524074	2.544438	-2.110895
H	3.312174	2.195396	-0.806729
H	5.298108	1.113963	0.136300
H	5.365486	-1.317200	0.524298
H	3.418212	-2.728910	-0.062236
H	1.051466	-3.691682	-0.881207
H	-1.244556	-4.040283	-1.750433
H	-2.567914	-2.109972	-2.532313
H	-1.690034	0.139976	-2.462102
H	-0.392245	1.837773	-3.520614
H	-0.543881	3.441033	-2.795928
H	-1.480164	2.099085	-2.150995
H	-0.671498	4.557837	1.516169
H	-1.695239	3.373196	2.343293
H	0.062174	3.247961	2.448173

1a (tg⁻) (-1230.325980 hartree)			
C	1.513609	-0.773743	3.191814
C	1.222779	-0.987486	1.707922
C	-0.165655	-0.471261	1.324144
C	-0.723867	-0.919055	-0.034077
C	0.127092	-0.601428	-1.262837
C	-3.227348	-1.162621	-0.140094
C	-4.349119	-0.321337	-0.260815
C	-3.841880	1.023686	-0.366876
C	-2.437900	0.932227	-0.305915
C	-4.456444	2.263744	-0.499353
C	-3.672569	3.398978	-0.565418
C	-2.280920	3.299122	-0.499137
C	-1.645700	2.078249	-0.369936
C	-3.376919	-2.541122	-0.010732
C	-4.660068	-3.056430	-0.006424
C	-5.780181	-2.230711	-0.127668
C	-5.629183	-0.863132	-0.255318
C	2.562944	0.894447	0.643073
C	3.594720	0.990876	-0.310825
C	3.974762	-0.357809	-0.647442
C	3.151998	-1.205435	0.115198
C	4.918266	-0.896883	-1.514186
C	5.031495	-2.270255	-1.613994
C	4.209643	-3.103562	-0.851994
C	3.265126	-2.590011	0.017874
C	1.995215	2.045266	1.189370
C	2.455856	3.272477	0.749036
C	3.467122	3.374989	-0.207969
C	4.043408	2.235956	-0.735410
N	2.290776	-0.442073	0.882589
N	-2.076065	-0.398181	-0.175831
H	0.773916	-1.292249	3.803002
H	1.486723	0.284622	3.453423
H	2.502567	-1.158755	3.441068
H	1.232152	-2.062597	1.521618
H	-0.188200	0.615179	1.380456
H	-0.864752	-0.824949	2.085684
H	-0.832763	-2.004001	0.001889
H	-0.458535	-0.790853	-2.162477
H	1.009777	-1.237718	-1.294260
H	0.464399	0.433512	-1.285508
H	-5.536129	2.335923	-0.548068
H	-4.135165	4.371800	-0.668672
H	-1.679536	4.198060	-0.551964
H	-0.566674	2.037532	-0.322046
H	-2.526628	-3.203541	0.082392
H	-4.796830	-4.125987	0.092326
H	-6.769912	-2.668089	-0.121363
H	-6.495748	-0.220034	-0.349049
H	5.553901	-0.247750	-2.103799
H	5.760541	-2.706282	-2.284135
H	4.314172	-4.177585	-0.939980
H	2.646524	-3.258703	0.601324
H	1.224416	2.003353	1.944583
H	2.022474	4.173573	1.164371
H	3.803363	4.351838	-0.529516
H	4.835299	2.309055	-1.470685

1a (g⁺g⁺) (-1230.317805 hartree)			
C	-0.320515	-0.418213	2.034754
C	0.663728	-1.200500	1.176349
C	-0.000048	-1.940658	0.000194
C	-0.663916	-1.200798	-1.176102
C	0.320233	-0.418490	-2.034597
C	-2.110575	0.895267	-0.647788
C	-3.110936	3.446728	-0.145842
C	-1.324007	2.008435	-0.943841
C	-3.402112	1.075946	-0.109116
C	-3.897952	2.349956	0.141089
C	-1.837326	3.265749	-0.685650
C	-3.003360	-1.147965	-0.378062
C	-5.458765	-2.043137	0.537052
C	-3.973086	-0.233752	0.062950
C	-3.261961	-2.517745	-0.375485
C	-4.493864	-2.945601	0.084550
C	-5.204150	-0.685326	0.523858
C	3.003028	-1.148131	0.378141
C	5.458077	-2.044067	-0.537159
C	3.261207	-2.517987	0.375572
C	3.972971	-0.234226	-0.063007
C	5.203865	-0.686180	-0.524001
C	4.492942	-2.946229	-0.084554
C	2.110820	0.895363	0.647812
C	3.111942	3.446512	0.145578
C	3.402375	1.075635	0.109019
C	1.324647	2.008828	0.943854
C	1.838331	3.265960	0.685520
C	3.898583	2.349475	-0.141334
N	-1.870271	-0.467406	-0.793783
N	1.870168	-0.467256	0.793946
H	-0.332856	1.923396	-1.353351
H	-4.891204	2.474728	0.554717
H	-1.224962	4.129760	-0.911775
H	-3.479911	4.446472	0.041953
H	-1.058208	-1.118624	2.427738
H	-0.866320	0.340937	1.482331
H	0.187013	0.051622	2.877122
H	1.032536	-2.003417	1.823409
H	-0.745511	-2.598199	0.452812
H	0.745507	-2.598193	-0.452291
H	-1.032703	-2.003826	-1.823041
H	1.058235	-1.118786	-2.427205
H	0.865650	0.341070	-1.482338
H	-0.187307	0.050876	-2.877218
H	-2.537456	-3.238530	-0.729124
H	-4.713248	-4.005870	0.090970
H	-5.953662	0.019055	0.863333
H	-6.411595	-2.412472	0.892838
H	2.536493	-3.238535	0.729273
H	5.953550	0.017966	-0.863580
H	4.712005	-4.006565	-0.090964
H	6.410765	-2.413699	-0.893018
H	0.333528	1.924175	1.353487
H	1.226261	4.130179	0.911643
H	4.891838	2.473904	-0.555061
H	3.481208	4.446128	-0.042327

1a (g^+g^-) (-1230.322422 hartree)			
C	-1.655353	-3.111231	1.740905
C	-1.275366	-1.636790	1.759613
C	0.184524	-1.379367	1.325396
C	0.421124	-1.130641	-0.170912
C	0.294142	-2.359412	-1.067004
C	-2.144182	0.617477	1.261014
C	-2.263526	3.380237	1.223712
C	-1.478540	1.330973	2.254809
C	-2.866948	1.281316	0.258521
C	-2.928812	2.670736	0.243310
C	-1.548336	2.710416	2.219286
C	-2.962566	-0.980620	-0.089807
C	-4.546884	-0.855352	-2.377815
C	-3.396716	0.258270	-0.604672
C	-3.340282	-2.164144	-0.726575
C	-4.123177	-2.080980	-1.862866
C	-4.187891	0.317398	-1.746051
C	1.781191	0.931740	-0.573286
C	2.494906	3.571889	-0.978935
C	0.773327	1.877769	-0.742646
C	3.137086	1.304052	-0.589841
C	3.491705	2.631978	-0.801147
C	1.150671	3.191990	-0.943405
C	2.958339	-0.938032	-0.177115
C	5.673280	-1.438711	0.075448
C	3.894102	0.103499	-0.337074
C	3.380640	-2.229652	0.132013
C	4.738589	-2.460740	0.252658
C	5.254339	-0.154730	-0.216034
N	1.683383	-0.433478	-0.356327
N	-2.207276	-0.754460	1.058396
H	-1.079355	-3.618819	2.515087
H	-1.419351	-3.595236	0.795549
H	-2.713347	-3.251952	1.961665
H	-1.341631	-1.331724	2.807065
H	0.803778	-2.210021	1.673153
H	0.546991	-0.488955	1.840158
H	-0.334978	-0.424178	-0.509558
H	-0.753719	-2.642696	-1.157294
H	0.846300	-3.217428	-0.685530
H	0.662704	-2.131198	-2.066891
H	-0.913146	0.836710	3.032834
H	-3.483455	3.186466	-0.530893
H	-1.029582	3.283046	2.977195
H	-3.052037	-3.134083	-0.357610
H	-4.416963	-2.996824	-2.360116
H	-4.515673	1.275333	-2.130720
H	-0.276478	1.619938	-0.704150
H	4.534579	2.924383	-0.818304
H	0.379624	3.941772	-1.068158
H	2.680652	-3.037875	0.283658
H	5.081935	-3.459387	0.492210
H	5.974855	0.644150	-0.342492
H	2.753198	4.610502	-1.139751
H	6.728870	-1.655325	0.174575
H	-5.160352	-0.828971	-3.268623
H	-2.292504	4.461885	1.223468

1a (g^-g^-) (-1230.322610 hartree)			
C	1.270030	0.212212	1.481031
C	-0.000060	0.000337	0.652188
C	-1.270117	-0.211782	1.481047
C	3.431795	-0.667060	0.560349
C	2.728394	1.280945	-0.307955
C	3.571011	-1.849904	1.282007
C	4.389113	-0.288138	-0.398056
C	2.029857	2.440270	-0.640802
C	3.938029	0.962468	-0.954705
C	4.679627	-2.636990	1.029287
C	5.495929	-1.095171	-0.633973
C	2.561756	3.264366	-1.614975
C	4.453913	1.807727	-1.930188
C	5.637366	-2.268436	0.081977
C	3.763476	2.958386	-2.256880
C	1.196630	1.388411	2.452321
C	-2.728218	-1.280854	-0.307928
C	-3.431973	0.667083	0.560221
C	-2.029547	-2.440157	-0.640563
C	-3.937850	-0.962587	-0.954793
C	-3.571460	1.849893	1.281879
C	-4.389167	0.287969	-0.398233
C	-2.561275	-3.264403	-1.614705
C	-4.453559	-1.807998	-1.930234
C	-4.680208	2.636769	1.029086
C	-5.496114	1.094799	-0.634229
C	-3.762960	-2.958605	-2.256762
C	-5.637811	2.268042	0.081704
C	-1.196548	-1.387983	2.452343
N	-2.434200	-0.290538	0.615088
N	2.434193	0.290762	0.615148
H	0.165321	-0.857839	-0.001528
H	-0.165529	0.858683	-0.001278
H	1.426086	-0.683999	2.080199
H	-1.426355	0.684414	2.080193
H	2.844650	-2.161510	2.020830
H	1.094945	2.702632	-0.167854
H	4.805156	-3.560566	1.580239
H	6.235798	-0.807209	-1.370730
H	2.029057	4.167186	-1.885776
H	5.384332	1.563298	-2.427920
H	6.492698	-2.908376	-0.090390
H	4.150486	3.626388	-3.015051
H	0.464463	1.190397	3.236558
H	0.912798	2.315947	1.955904
H	2.165193	1.540080	2.928182
H	-1.094690	-2.702407	-0.167446
H	-2.845215	2.161622	2.020766
H	-2.028474	-4.167209	-1.885352
H	-5.383972	-1.563734	-2.428059
H	-4.805952	3.560317	1.580038
H	-6.235885	0.806692	-1.371029
H	-4.149825	-3.626716	-3.014910
H	-6.493251	2.907823	-0.090720
H	-0.912005	-2.315365	1.956034
H	-2.165249	-1.540161	2.927760
H	-0.464852	-1.189624	3.236928

1b (tt #1) (-1859.418107 hartree)

C	-2.087601	-0.437485	1.286985
C	-0.123914	-0.780423	-1.676100
C	-1.832995	-1.816853	1.168662
C	0.122671	-0.838009	1.647754
C	4.381177	-0.696233	-0.672952
C	2.082160	-0.369139	-1.307166
C	0.820299	0.251991	-1.613031
C	1.457434	-0.551562	1.911374
C	0.429226	1.575677	-1.788681
C	1.808472	0.767481	2.094885
C	-0.204570	-3.259007	-1.354257
C	3.344313	0.145459	-1.037275
C	1.839362	-1.748393	-1.210749
C	-5.777024	-0.207914	0.373382
C	-4.394436	-0.790465	0.680856
C	-3.355733	0.060111	1.040337
C	-0.827441	0.189432	1.598505
C	0.014987	-4.062686	-0.060389
C	-0.442414	1.512600	1.789399
C	-4.119647	-2.161055	0.598397
C	-0.898220	1.883969	-2.026807
C	0.884056	1.823860	2.033923
C	-1.399100	3.318363	-2.203795
C	2.877431	-2.617615	-0.885274
C	-1.819475	0.824195	-2.095027
C	-1.461555	-0.495669	-1.930054
C	5.775298	-0.181644	-0.308819
C	-2.861565	-2.689375	0.839797
C	4.119176	-2.076622	-0.618552
C	0.228264	-3.305494	1.261997
C	-0.269031	4.343393	-2.094073
C	-2.051144	3.471670	-3.586029
C	-2.438120	3.631722	-1.116602
C	1.382414	3.257293	2.225317
C	-0.045838	-4.216019	2.455645
C	0.083196	-4.122091	-2.579553
C	-5.666519	0.743891	-0.827517
C	-6.294030	0.570856	1.592033
C	0.252465	4.282279	2.116572
C	5.881724	1.338970	-0.440075
C	6.086569	-0.556376	1.148519
C	6.824496	-0.815159	-1.234425
C	2.025524	3.400299	3.612869
C	2.427478	3.579585	1.146284
C	-6.804318	-1.288756	0.029688
N	0.498160	-1.992689	-1.426546
N	-0.489261	-2.051406	1.379389
H	2.221731	-1.316977	1.933301
H	1.178019	2.353003	-1.724508
H	2.853567	0.986824	2.275769
H	-1.259074	-2.989704	-1.372016
H	3.487691	1.216276	-1.087487
H	-3.518746	1.129643	1.105253
H	-0.843744	-4.732361	0.032685
H	0.878367	-4.722503	-0.178063
H	-1.193661	2.288141	1.730600
H	-4.909084	-2.851491	0.336679
H	2.730112	-3.687560	-0.828139
H	-2.866457	1.041389	-2.266488
H	-2.222579	-1.264347	-1.957666
H	-2.702375	-3.756337	0.762722
H	4.919004	-2.756920	-0.352673
H	1.279513	-3.024316	1.288154
H	0.217747	4.306085	-1.117839
H	-0.672172	5.349719	-2.223333
H	0.491361	4.189283	-2.862266
H	-1.331473	3.254374	-4.377666
H	-2.415483	4.492796	-3.724063
H	-2.898237	2.795352	-3.708627
H	-3.302248	2.970010	-1.187313
H	-2.795216	4.660054	-1.211645
H	-2.004080	3.509254	-0.122705
H	0.224100	-3.714183	3.384962
H	0.534432	-5.137224	2.376825
H	-1.103257	-4.481747	2.506487
H	-0.189967	-3.589863	-3.490849
H	-0.486175	-5.052398	-2.536268
H	1.144050	-4.373024	-2.636114
H	-4.982760	1.568076	-0.619943
H	-6.642606	1.171280	-1.069942
H	-5.296708	0.213363	-1.707760
H	-6.377705	-0.084106	2.461377
H	-7.280515	0.992340	1.383661
H	-5.629131	1.394092	1.856397
H	-0.228071	4.251220	1.137192
H	0.653948	5.288018	2.255248
H	-0.512330	4.122214	2.879172
H	5.685134	1.671346	-1.461237
H	6.890862	1.659706	-0.174487
H	5.186039	1.852050	0.227134
H	5.350358	-0.115817	1.824536
H	7.075720	-0.190434	1.434641
H	6.073474	-1.636964	1.298517
H	6.835467	-1.902695	-1.149970
H	7.823527	-0.451650	-0.981515
H	6.621682	-0.562168	-2.276825
H	1.300994	3.176099	4.398115
H	2.387931	4.420642	3.761264
H	2.872426	2.723865	3.736184
H	3.287233	2.910723	1.207525
H	2.791020	4.604021	1.257436
H	1.995362	3.474590	0.149561
H	-6.516088	-1.854270	-0.858793
H	-7.769765	-0.822280	-0.175656
H	-6.944453	-1.990137	0.854688

1b (tt #2) (-1859.406276 hartree)

C	1.360653	-0.377320	-1.161687
C	5.679916	-1.295626	0.278130
C	2.976117	-1.739943	-0.384211
C	4.893283	-0.252297	-0.210022
C	-1.304772	0.118945	1.260007
C	-3.274896	-0.670757	0.411624
C	-2.587116	0.523112	0.846221
C	3.557600	-0.462209	-0.537181
C	7.150815	-1.035498	0.632734
C	-4.984008	-2.169203	-0.412514
C	2.518286	0.409147	-1.036491
C	-2.104684	2.809045	1.471652
C	0.428181	-3.165897	0.771331
C	0.663508	-2.755454	-0.686249
C	-2.977552	1.854833	0.952603
C	-0.075523	-2.029025	1.671245
C	0.164594	0.192827	-1.600980
C	-2.609540	4.239167	1.721835
C	-6.375982	-2.467728	-0.983646
C	-4.554102	-0.883859	-0.102544
C	5.076478	-2.556089	0.423456
C	2.492193	1.766043	-1.362645
C	-2.387096	-1.746235	0.614191
C	-2.809637	-3.048119	0.326090
C	3.745550	-2.797286	0.102032
C	1.313070	2.356482	-1.798318
C	0.163029	1.544359	-1.896770
C	-4.087707	-3.230704	-0.178606
C	-0.811581	2.383887	1.832075
C	1.209061	3.853131	-2.112810
C	-0.395707	1.061310	1.735984
C	-1.494086	5.185661	2.186544
C	-3.241667	4.832795	0.449412
C	1.012192	-3.940525	-1.588728
C	-0.444584	-2.537105	3.066626
C	-3.680105	4.183651	2.830784
C	7.231314	0.045703	1.727218
C	7.902448	-0.550519	-0.621516
C	2.581321	4.540407	-2.141023
C	0.544746	4.069320	-3.485086
C	0.357034	4.513465	-1.015474
C	-7.211385	-1.193408	-1.167847
C	7.858399	-2.293845	1.154485
C	-7.136698	-3.403794	-0.025316
C	-6.232022	-3.148064	-2.358380
N	-1.172712	-1.258916	1.091207
N	1.643789	-1.683576	-0.776078
H	5.315361	0.741272	-0.339309
H	-0.258225	-4.017127	0.793705
H	1.361988	-3.528654	1.214067
H	-0.271374	-2.332388	-1.060492
H	0.753410	-1.321068	1.762190
H	-0.754683	-0.378488	-1.680367
H	-5.203828	-0.027517	-0.249515
H	5.657608	-3.390864	0.799039
H	3.404235	2.345346	-1.263872
H	-2.180080	-3.911875	0.495857
H	3.340476	-3.794340	0.231082
H	-0.779638	1.989569	-2.200474
H	-4.400374	-4.247966	-0.397770
H	-0.097076	3.105986	2.211347
H	0.616262	0.792974	2.020968
H	-0.690252	5.260661	1.447402
H	-1.060041	4.865195	3.139943
H	-1.904355	6.190585	2.335448
H	-3.629208	5.837386	0.656224
H	-4.077792	4.227887	0.084800
H	-2.507279	4.914402	-0.357657
H	1.176057	-3.595029	-2.613146
H	0.184696	-4.657618	-1.593364
H	1.912991	-4.465539	-1.259854
H	-0.722249	-1.699971	3.713737
H	0.399871	-3.064590	3.522415
H	-1.294806	-3.225657	3.013768
H	-3.262872	3.760623	3.751069
H	-4.531636	3.562563	2.533366
H	-4.056010	5.189723	3.053635
H	6.787045	0.990306	1.397094
H	8.276427	0.243453	1.994179
H	6.701357	-0.276175	2.630325
H	7.857227	-1.303258	-1.415982
H	8.956809	-0.360981	-0.386804
H	7.476238	0.377673	-1.015728
H	3.078301	4.499776	-1.165644
H	2.458631	5.597742	-2.400522
H	3.244262	4.086112	-2.886007
H	1.118024	3.574860	-4.276773
H	0.493419	5.139881	-3.716101
H	-0.476900	3.678461	-3.515413
H	-0.622120	4.032199	-0.933250
H	0.205621	5.580868	-1.220082
H	0.849971	4.412359	-0.042189
H	-8.193188	-1.452739	-1.578715
H	-6.735258	-0.493782	-1.863795
H	-7.376912	-0.676233	-0.216199
H	7.388405	-2.674885	2.068048
H	8.901036	-2.057087	1.393201
H	7.864125	-3.095640	0.407742
H	-7.249353	-2.940592	0.960873
H	-6.616702	-4.357742	0.109931
H	-8.136657	-3.622449	-0.418619
H	-7.219553	-3.364443	-2.783095
H	-5.685226	-4.094078	-2.288641
H	-5.692567	-2.499645	-3.057287
H	-3.987180	2.131327	0.659894

1b (tg⁺) (-1859.415314 hartree)							
C	-0.681049	3.962589	-2.045670	H	-0.705049	5.053251	-2.065436
C	0.578864	3.520292	-1.317186	H	-1.590688	3.622421	-1.554550
C	0.610968	3.989346	0.148241	H	-0.687163	3.602290	-3.073746
C	-0.577274	3.656585	1.057345	H	1.406668	4.045318	-1.804502
C	-0.354969	4.211648	2.463586	H	0.686287	5.080973	0.132393
C	2.191204	1.650847	-1.242747	H	1.534913	3.636384	0.605209
C	4.577963	0.244849	-0.815579	H	-1.452326	4.156868	0.642975
C	3.348497	2.359217	-0.933756	H	-0.177095	5.287692	2.429119
C	2.227631	0.255759	-1.334685	H	0.505140	3.738310	2.939068
C	3.414381	-0.436680	-1.133761	H	-1.231299	4.024290	3.084490
C	4.511140	1.644437	-0.718856	H	3.358482	3.438216	-0.851697
C	0.095016	0.998997	-1.707458	H	3.409108	-1.514313	-1.218961
C	-1.043774	-1.556894	-2.016504	H	5.405110	2.201372	-0.467042
C	0.881750	-0.165804	-1.620535	H	-1.916793	1.726910	-2.070657
C	-1.259281	0.878430	-1.993191	H	-2.860632	-0.446718	-2.329424
C	-1.798393	-0.386796	-2.139756	H	0.947087	-2.294304	-1.664046
C	0.313819	-1.420468	-1.766181	H	-3.384848	3.468920	0.267948
C	-2.208690	1.760848	0.910448	H	-3.382968	-1.410004	1.208962
C	-4.556186	0.269939	0.556946	H	-5.396167	2.159112	-0.043890
C	-3.361829	2.406860	0.476269	H	1.852970	2.027408	1.828785
C	-2.232733	0.385487	1.177016	H	2.834295	-0.093736	2.399554
C	-3.399723	-0.348531	1.001274	H	-0.932295	-2.080389	1.921579
C	-4.504263	1.651157	0.301773	H	7.215746	-0.790927	1.164994
C	-0.129535	1.195297	1.523079	H	5.496691	-0.732003	1.580858
C	1.032295	-1.260662	2.216245	H	6.365714	0.758126	1.206815
C	-0.895740	0.021642	1.564748	H	7.961213	-0.389033	-1.230620
C	1.222696	1.151322	1.850796	H	7.180930	1.189555	-1.197503
C	1.774238	-0.068341	2.179724	H	6.773610	0.038712	-2.471420
C	-0.314543	-1.193304	1.909229	H	6.804432	-2.424674	-0.667113
C	5.904310	-0.468689	-0.541809	H	5.566620	-2.147518	-1.893589
C	6.267434	-0.295852	0.941200	H	5.104159	-2.479905	-0.217834
C	7.018418	0.132703	-1.411704	H	-1.995154	-3.115334	0.025912
C	5.830408	-1.966467	-0.849580	H	-0.437960	-3.756552	-0.497255
C	-1.664116	-2.951872	-2.115181	H	-1.915428	-4.676275	-0.809773
C	-1.491276	-3.669834	-0.768656	H	-3.709727	-2.359273	-1.662204
C	-3.158557	-2.898221	-2.434587	H	-3.562055	-3.911509	-2.485415
C	-0.964835	-3.760318	-3.217153	H	-3.350837	-2.413114	-3.393737
C	-5.867832	-0.486298	0.333142	H	0.100162	-3.881793	-3.014670
C	-5.759313	-1.961669	0.725308	H	-1.067122	-3.263315	-4.183719
C	-6.981989	0.145125	1.182040	H	-1.405184	-4.757522	-3.295449
C	-6.254658	-0.407412	-1.151663	H	-5.009649	-2.489510	0.133153
C	1.736894	-2.574456	2.557979	H	-6.717851	-2.455954	0.555995
C	2.752668	-2.889516	1.451469	H	-5.506579	-2.078826	1.781047
C	2.472158	-2.445599	3.900133	H	-7.155496	1.188081	0.913975
C	0.759269	-3.746314	2.664928	H	-6.725777	0.109738	2.242661
N	-0.928136	2.248789	1.104170	H	-7.920640	-0.395460	1.037888
N	0.903740	2.107556	-1.471909	H	-5.482911	-0.858321	-1.778763
				H	-6.388073	0.625611	-1.476609
				H	-7.192326	-0.939280	-1.330932
				H	3.481893	-2.085112	1.347385
				H	2.247848	-3.002439	0.489113
				H	3.291562	-3.814851	1.671024
				H	3.236505	-1.668017	3.872785
				H	2.966421	-3.386977	4.152270
				H	1.772682	-2.200607	4.701711
				H	0.239461	-3.929311	1.723000
				H	0.011080	-3.575251	3.441714
				H	1.305320	-4.656121	2.922056

1b (tg^{-}) (-1859.398778 hartree)

C	1.310359	-1.897736	3.944806
C	1.210897	-1.953936	2.422571
C	-0.169377	-1.501309	1.928290
C	-0.605279	-1.936597	0.515724
C	0.450266	-1.881097	-0.586394
C	-3.075940	-1.699618	0.060163
C	-3.973552	-0.666680	-0.267158
C	-3.187722	0.538793	-0.376172
C	-1.858680	0.173567	-0.115347
C	-3.514745	1.863564	-0.654145
C	-2.532645	2.840164	-0.665043
C	-1.212335	2.443965	-0.394304
C	-0.855409	1.139380	-0.129471
C	-3.539772	-2.994406	0.245493
C	-4.896144	-3.233630	0.096967
C	-5.813249	-2.227376	-0.231435
C	-5.325493	-0.938428	-0.410574
C	2.356570	0.169553	1.632939
C	3.320695	0.483840	0.658852
C	3.921161	-0.766701	0.256298
C	3.267195	-1.772983	0.985133
C	4.923895	-1.093911	-0.642283
C	5.296808	-2.418999	-0.835552
C	4.631019	-3.399826	-0.090892
C	3.626035	-3.101823	0.815368
C	1.654131	1.189780	2.261173
C	1.861425	2.491893	1.839305
C	2.760599	2.824659	0.820424
C	3.508741	1.797345	0.257232
C	-2.826238	4.316023	-0.942070
C	-2.461502	5.147748	0.297159
C	-1.988430	4.795065	-2.137511
C	-4.301335	4.561414	-1.265318
C	-7.311837	-2.499377	-0.397003
C	-7.668514	-3.968426	-0.158433
C	-8.101806	-1.648715	0.609279
C	-7.744421	-2.130289	-1.824215
C	2.932889	4.258087	0.310416
C	1.998431	5.242515	1.016814
C	4.379888	4.718478	0.539454
C	2.615877	4.301013	-1.192742
C	6.405841	-2.754691	-1.837242
C	7.706356	-2.054033	-1.415421
C	5.995600	-2.267320	-3.235279
C	6.681526	-4.257696	-1.920464
N	2.297828	-1.207585	1.797105
N	-1.795149	-1.186583	0.142591

H	0.575976	-2.563866	4.399793
H	1.126663	-0.888952	4.316842
H	2.305579	-2.201839	4.269569
H	1.356840	-2.992332	2.118660
H	-0.233328	-0.417230	1.991173
H	-0.919085	-1.882703	2.625261
H	-0.925324	-2.977520	0.580173
H	-0.026235	-2.097418	-1.542763
H	1.222545	-2.629834	-0.417498
H	0.941074	-0.912700	-0.662524
H	-4.548046	2.113831	-0.853631
H	-0.422832	3.186032	-0.383918
H	0.178008	0.906301	0.075137
H	-2.876059	-3.810808	0.499073
H	-5.242542	-4.246931	0.243793
H	-5.999020	-0.128149	-0.664372
H	5.414085	-0.300844	-1.194664
H	4.902623	-4.439036	-0.211207
H	3.150645	-3.899116	1.371951
H	0.940539	0.996003	3.048749
H	1.281589	3.267128	2.319346
H	4.239130	2.010412	-0.514425
H	-1.405167	5.046174	0.550915
H	-2.661752	6.207151	0.118478
H	-3.045865	4.828823	1.162449
H	-2.197836	5.846250	-2.350454
H	-0.918430	4.702590	-1.944846
H	-2.219782	4.211517	-3.030651
H	-4.617385	4.005456	-2.150383
H	-4.949629	4.281487	-0.432551
H	-4.460956	5.622770	-1.465228
H	-7.162780	-4.628041	-0.866465
H	-7.413028	-4.287771	0.853987
H	-8.743616	-4.107994	-0.287306
H	-7.929322	-0.582246	0.457972
H	-9.174006	-1.832201	0.503793
H	-7.811452	-1.892562	1.633089
H	-7.562650	-1.076537	-2.040050
H	-7.195153	-2.721698	-2.559350
H	-8.812150	-2.320148	-1.959594
H	0.948848	4.976225	0.873232
H	2.195644	5.293512	2.089320
H	2.143871	6.243251	0.606000
H	5.092648	4.078627	0.017116
H	4.516883	5.739254	0.174366
H	4.628161	4.698414	1.602343
H	2.717694	5.319873	-1.573963
H	3.287673	3.660970	-1.766118
H	1.594445	3.965004	-1.384627
H	8.508533	-2.285048	-2.120579
H	7.589132	-0.969812	-1.387226
H	8.018124	-2.382776	-0.422148
H	6.776666	-2.499704	-3.963285
H	5.071703	-2.751332	-3.557815
H	5.832261	-1.188888	-3.254372
H	7.012181	-4.662755	-0.961940
H	5.799821	-4.813922	-2.245159
H	7.474412	-4.442627	-2.647660

1b (g^+g^+) (-1859.393307 hartree)

C	1.763873	0.215366	1.002043
C	3.064921	0.525061	0.567057
C	3.474518	1.843557	0.394643
C	2.605439	2.886303	0.659954
C	1.321881	2.555355	1.120217
C	0.887616	1.258927	1.296814
C	2.842315	-1.732770	0.724839
C	3.758271	-0.724703	0.390963
C	3.232291	-3.065605	0.684764
C	4.529469	-3.360529	0.298892
C	5.459860	-2.374487	-0.052391
C	5.050320	-1.048273	0.006019
C	6.891371	-2.707295	-0.484281
C	7.163410	-4.213296	-0.492895
C	7.134966	-2.171102	-1.903209
C	7.884901	-2.049305	0.485393
C	2.977134	4.354814	0.448689
C	4.437461	4.525817	0.025270
C	2.085638	4.943272	-0.655846
C	2.760876	5.142834	1.748995
C	0.448917	-2.010696	1.274215
C	-0.000002	-2.753136	0.000132
C	-0.448935	-2.010791	-1.274001
C	0.700900	-1.334420	-2.008281
C	-0.700924	-1.334266	2.008430
C	-2.842358	-1.732775	-0.724736
C	-3.758299	-0.724665	-0.390946
C	-3.064892	0.525069	-0.567033
C	-1.763830	0.215314	-1.001932
C	-3.474452	1.843586	-0.394687
C	-2.605316	2.886292	-0.659974
C	-1.321738	2.555283	-1.120135
C	-0.887511	1.258833	-1.296668
C	-3.232392	-3.065594	-0.684660
C	-4.529606	-3.360458	-0.298863
C	-5.459979	-2.374371	0.052344
C	-5.050383	-1.048175	-0.006073
C	-2.976969	4.354821	-0.448763
C	-4.437334	4.525892	-0.025500
C	-2.085561	4.943245	0.655863
C	-2.760539	5.142825	-1.749049
C	-6.891530	-2.707113	0.484149
C	-7.163631	-4.213102	0.492770
C	-7.135194	-2.170887	1.903052
C	-7.884971	-2.049098	-0.485599
N	1.626463	-1.168985	1.076513
N	-1.626462	-1.169043	-1.076345
H	4.482777	2.031386	0.050875
H	0.613093	3.345779	1.333881
H	-0.121628	1.093182	1.631814
H	2.560498	-3.868918	0.955919
H	4.817915	-4.401922	0.276169
H	5.738714	-0.249929	-0.245448
H	8.192313	-4.395844	-0.808880
H	7.039983	-4.652868	0.498900
H	6.506122	-4.739603	-1.188049
H	8.152935	-2.401157	-2.227251
H	7.005585	-1.089044	-1.952574
H	6.439279	-2.624084	-2.612328
H	7.729970	-2.412916	1.503042
H	8.912008	-2.279386	0.191565
H	7.776640	-0.963913	0.498147
H	4.660328	5.587111	-0.100354
H	5.123232	4.128588	0.776420
H	4.643884	4.031353	-0.926050
H	2.336090	5.992003	-0.833446
H	1.030546	4.894818	-0.382716
H	2.216433	4.395689	-1.591541
H	3.377343	4.737459	2.553562
H	3.031175	6.191744	1.605159
H	1.720403	5.110623	2.074843
H	0.790871	-2.808247	1.943333
H	0.809875	-3.411184	-0.320679
H	-0.809864	-3.411180	0.320986
H	-0.790904	-2.808387	-1.943059
H	0.352304	-0.870137	-2.930786
H	1.435681	-2.097252	-2.266956
H	1.213468	-0.588108	-1.407533
H	-1.435703	-2.097079	2.267169
H	-0.352339	-0.869898	2.930896
H	-1.213492	-0.588014	1.407609
H	-4.482728	2.031464	-0.050994
H	-0.612903	3.345674	-1.333765
H	0.121754	1.093038	-1.631580
H	-2.560618	-3.868940	-0.955761
H	-4.818097	-4.401838	-0.276138
H	-5.738763	-0.249798	0.245327
H	-4.660171	5.587198	0.100075
H	-5.123041	4.128670	-0.776713
H	-4.643877	4.031460	0.925811
H	-2.335981	5.991989	0.833427
H	-1.030443	4.894736	0.382844
H	-2.216483	4.395676	1.591549
H	-1.720033	5.110572	-2.074787
H	-3.376937	4.737475	-2.553681
H	-3.030809	6.191747	-1.605246
H	-8.192560	-4.395603	0.808696
H	-7.040162	-4.652696	-0.499010
H	-6.506406	-4.739424	1.187973
H	-8.153193	-2.400896	2.227033
H	-7.005774	-1.088833	1.952409
H	-6.439571	-2.623886	2.612223
H	-7.729991	-2.412733	-1.503232
H	-8.912107	-2.279130	-0.191833
H	-7.776663	-0.963710	-0.498365

1b (g$^+$g$^-$) (-1859.397172 hartree)

C	1.195248	-3.603759	-2.845837
C	0.993014	-2.115638	-2.593766
C	-0.371980	-1.787163	-1.952310
C	-0.426467	-1.777874	-0.418867
C	-0.312787	-3.145403	0.249464
C	2.199603	-0.082224	-1.877136
C	2.784262	2.632363	-1.496941
C	1.537432	0.867889	-2.649999
C	3.128146	0.322339	-0.913131
C	3.420451	1.670860	-0.730723
C	1.839470	2.197793	-2.443254
C	2.945078	-1.959879	-0.900121
C	4.825715	-2.379557	1.160962
C	3.608327	-0.880631	-0.283201
C	3.242065	-3.255576	-0.488032
C	4.165047	-3.438058	0.528053
C	4.531603	-1.094500	0.728712
C	-1.571278	0.288198	0.403192
C	-2.054853	2.921014	1.225413
C	-0.494111	1.160783	0.536011
C	-2.876874	0.730850	0.656541
C	-3.110215	2.038456	1.073590
C	-0.759783	2.451977	0.941291
C	-2.923026	-1.442709	-0.037974
C	-5.715774	-1.732185	0.042650
C	-3.747204	-0.382331	0.373545
C	-3.495398	-2.649507	-0.434674
C	-4.869156	-2.768967	-0.385917
C	-5.129796	-0.535525	0.417947
C	3.078789	4.128362	-1.360591
C	1.813165	4.862824	-0.897574
C	3.525507	4.688708	-2.719777
C	4.188570	4.409050	-0.345171
C	5.841500	-2.591313	2.286833
C	5.356490	-1.870137	3.553672
C	6.034628	-4.070950	2.626298
C	7.201898	-2.016731	1.863543
C	-2.248130	4.375542	1.660948
C	-3.705921	4.686411	2.007153
C	-1.392881	4.664736	2.903781
C	-1.817076	5.309270	0.519768
C	-7.228980	-1.960530	0.068482
C	-7.723445	-2.293636	-1.347478
C	-7.555789	-3.129882	1.009861
C	-7.991177	-0.728024	0.559878
N	-1.600392	-1.041832	0.018731
N	2.091528	-1.465788	-1.883669

H	0.483799	-3.918864	-3.609637
H	1.008031	-4.209294	-1.961628
H	2.199974	-3.810548	-3.214226
H	0.987015	-1.648626	-3.582419
H	-1.122171	-2.476174	-2.349448
H	-0.672646	-0.787802	-2.270040
H	0.418425	-1.188886	-0.063037
H	0.717450	-3.494772	0.201715
H	-0.947275	-3.896287	-0.219949
H	-0.589677	-3.070912	1.300897
H	0.801888	0.592362	-3.393559
H	4.153628	1.949517	0.013836
H	1.317373	2.931811	-3.044021
H	2.786572	-4.123316	-0.935706
H	4.372789	-4.455414	0.828426
H	5.019392	-0.237627	1.178429
H	0.524447	0.867026	0.316239
H	-4.127572	2.349896	1.268245
H	0.080488	3.127300	1.035453
H	-2.898840	-3.479554	-0.783163
H	-5.302906	-3.711298	-0.697487
H	-5.733792	0.300389	0.743956
H	1.988693	5.940578	-0.856882
H	1.524570	4.530217	0.101097
H	0.969691	4.682852	-1.565499
H	3.761980	5.751636	-2.630779
H	2.748092	4.584445	-3.477562
H	4.415728	4.168465	-3.078423
H	4.367948	5.484360	-0.288855
H	5.127408	3.930036	-0.630447
H	3.917165	4.068569	0.656145
H	4.388753	-2.260438	3.874613
H	5.246325	-0.797338	3.389276
H	6.070617	-2.011833	4.368538
H	5.104989	-4.533294	2.964245
H	6.763691	-4.167588	3.433023
H	6.411061	-4.636817	1.771721
H	7.140729	-0.947372	1.656614
H	7.567229	-2.511886	0.961754
H	7.939108	-2.161778	2.656985
H	-4.363747	4.554440	1.145808
H	-3.791294	5.725944	2.329752
H	-4.070689	4.054622	2.819653
H	-0.329723	4.519935	2.707289
H	-1.673932	4.006787	3.728393
H	-1.532644	5.699152	3.227260
H	-2.414423	5.124700	-0.375395
H	-0.768675	5.164176	0.257901
H	-1.949226	6.354412	0.811169
H	-7.251659	-3.196434	-1.737977
H	-7.502315	-1.475493	-2.035735
H	-8.804070	-2.456547	-1.344228
H	-7.078725	-4.055447	0.684531
H	-8.634397	-3.303063	1.040677
H	-7.213957	-2.915666	2.024318
H	-7.825679	0.134605	-0.088723
H	-7.703922	-0.454706	1.577185
H	-9.062481	-0.938249	0.563584

1b (g$^-$g$^-$) (-1859.397748 hartree)

C	-0.640499	1.024343	-2.467826
C	0.478832	0.764403	-1.457396
C	1.809837	0.284664	-2.050153
C	-2.570467	-0.117335	-1.436957
C	-2.463597	2.081442	-1.006866
C	-2.335749	-1.419860	-1.856654
C	-3.585635	0.150334	-0.506313
C	-2.135938	3.427205	-0.907877
C	-3.521783	1.566833	-0.234085
C	-3.138012	-2.428507	-1.347529
C	-4.379021	-0.877303	-0.021646
C	-2.879115	4.228757	-0.056054
C	-4.252600	2.394240	0.602778
C	-4.166387	-2.189346	-0.429437
C	-3.946241	3.745176	0.708156
C	-0.377935	2.222981	-3.370303
C	2.485852	-1.222819	-0.128347
C	3.933846	0.395775	-0.700288
C	1.401222	-2.085693	-0.024699
C	3.601448	-1.360979	0.717092
C	4.593733	1.470451	-1.277398
C	4.532893	-0.321131	0.350262
C	1.409717	-3.093768	0.931629
C	3.611992	-2.372943	1.664221
C	5.845253	1.811404	-0.789140
C	5.785135	0.044270	0.817877
C	2.526678	-3.225298	1.766331
C	6.466134	1.118819	0.256665
C	1.681494	-0.794721	-3.126005
C	7.849235	1.501353	0.792916
C	8.446257	2.705850	0.061518
C	8.810820	0.315518	0.621421
C	7.740302	1.856011	2.283784
C	0.176064	-3.993853	1.056324
C	-0.158195	-4.620426	-0.306051
C	-1.013991	-3.143145	1.527057
C	0.382370	-5.129736	2.060613
C	-5.058005	-3.310142	0.113476
C	-6.478879	-3.129285	-0.442135
C	-5.106205	-3.254241	1.647877
C	-4.553102	-4.697936	-0.289496
C	-4.765328	4.638644	1.644144
C	-4.287189	6.092280	1.628679
C	-6.239245	4.616768	1.211476
C	-4.649773	4.113604	3.083339
N	-1.920670	1.059874	-1.777154
N	2.692917	-0.149683	-0.981602

H	0.113759	0.017800	-0.753842
H	0.660648	1.670611	-0.873385
H	-0.718087	0.158360	-3.124899
H	2.303041	1.140137	-2.512515
H	-1.548351	-1.666735	-2.557315
H	-1.325368	3.864815	-1.469175
H	-2.934220	-3.438514	-1.673260
H	-5.162168	-0.644547	0.689794
H	-2.605468	5.272553	0.006697
H	-5.063468	1.968387	1.182093
H	0.402366	1.967759	-4.089503
H	-0.029252	3.091008	-2.812520
H	-1.273576	2.498984	-3.926391
H	0.541316	-1.976393	-0.669027
H	4.161063	2.043071	-2.087409
H	4.462378	-2.497508	2.323805
H	6.347271	2.651259	-1.248223
H	6.227532	-0.521323	1.629495
H	2.554331	-4.004998	2.513389
H	1.098750	-1.649878	-2.782231
H	2.673183	-1.153020	-3.401704
H	1.207061	-0.398085	-4.024480
H	7.822415	3.595355	0.170201
H	9.427233	2.937213	0.481040
H	8.581337	2.506800	-1.003541
H	9.803641	0.573115	0.998592
H	8.465384	-0.564916	1.165063
H	8.904364	0.043063	-0.431638
H	8.719775	2.133282	2.681342
H	7.059663	2.696580	2.432281
H	7.367268	1.016165	2.871567
H	-0.395875	-3.863059	-1.053857
H	0.678642	-5.215301	-0.677215
H	-1.029229	-5.274113	-0.214743
H	-1.249551	-2.346187	0.820603
H	-1.904871	-3.768236	1.625796
H	-0.804972	-2.688601	2.497374
H	1.223191	-5.768085	1.781584
H	0.556062	-4.751769	3.069670
H	-0.513498	-5.752902	2.093730
H	-7.139809	-3.915114	-0.067663
H	-6.898622	-2.165810	-0.147776
H	-6.476240	-3.173910	-1.532973
H	-5.721391	-4.070234	2.034091
H	-4.106091	-3.348120	2.074079
H	-5.536623	-2.320064	2.009878
H	-4.571126	-4.840876	-1.371494
H	-3.533952	-4.871741	0.063270
H	-5.194781	-5.462715	0.151985
H	-3.251066	6.181394	1.961286
H	-4.369877	6.534907	0.633960
H	-4.903492	6.685804	2.306672
H	-6.655325	3.609045	1.249657
H	-6.838156	5.250757	1.869896
H	-6.347733	4.985065	0.189473
H	-3.609277	4.116992	3.413728
H	-5.228211	4.742162	3.764879
H	-5.025014	3.093018	3.171214

Table S5. Optimized Geometries of Carbazole Dimers **1a** and **1b** in the Excited (S_1) State at the ωB97X-D/def2-TZVP Level.

1a (tt) (-1227.868305 hartree)			
C	0.338560	-3.220087	1.684132
C	-0.654052	-2.337069	2.117599
C	-0.514065	-0.950442	1.944536
C	0.671749	-0.490191	1.349635
C	1.693600	-1.382273	0.917918
C	1.523842	-2.761413	1.081098
C	2.327498	0.782458	0.475571
C	3.135324	1.833245	0.047476
C	4.382379	1.515935	-0.521050
C	4.797056	0.186426	-0.642602
C	3.992256	-0.868346	-0.203254
C	2.740908	-0.573616	0.357879
C	0.233341	1.977757	1.278439
C	0.000063	2.798009	-0.000091
C	-0.233161	1.977588	-1.278528
C	-2.327413	0.782512	-0.475607
C	-0.671754	-0.490352	-1.349524
C	-3.135139	1.833407	-0.047586
C	-2.740972	-0.573511	-0.357867
C	0.514039	-0.950763	-1.944350
C	-1.693727	-1.382307	-0.917833
C	-4.382250	1.516259	0.520901
C	-3.992375	-0.868078	0.203222
C	0.653864	-2.337409	-2.117375
C	-1.524122	-2.761470	-1.080958
C	-4.797079	0.186800	0.642492
C	-0.338867	-3.220301	-1.683925
C	0.760361	2.834550	2.426432
C	-0.760068	2.834257	-2.426665
N	1.058126	0.801112	1.054455
N	-1.058024	0.801011	-1.054453
H	0.194129	-4.294085	1.826450
H	-1.560341	-2.726120	2.585632
H	-1.292363	-0.269727	2.290978
H	2.293911	-3.463993	0.758840
H	2.836256	2.875577	0.155024
H	5.034948	2.322747	-0.860949
H	5.773559	-0.030751	-1.083041
H	4.331007	-1.902592	-0.291845
H	-0.851282	3.466375	0.199408
H	0.851375	3.466391	-0.199670
H	-2.835946	2.875702	-0.155163
H	1.292429	-0.270152	-2.290789
H	-5.034745	2.323157	0.860739
H	-4.331243	-1.902284	0.291845
H	1.560129	-2.726579	-2.585357
H	-2.294288	-3.463951	-0.758715
H	-5.773625	-0.030251	1.082898
H	-0.194555	-4.294320	-1.826206
H	1.759572	3.239352	2.209737
H	0.826617	2.239557	3.348527
H	0.079714	3.679241	2.610846
H	-0.747262	1.582733	1.577888
H	0.747441	1.582482	-1.577869
H	-0.079368	3.678891	-2.611144
H	-1.759272	3.239136	-2.210086
H	-0.826293	2.239149	-3.348688

1a (tg^+) (-1227.854777 hartree)			
C	1.027918	-1.180304	2.688409
C	0.252673	-2.333070	2.557529
C	-0.975739	-2.326927	1.880064
C	-1.436542	-1.117849	1.314959
C	-0.634767	0.058756	1.454492
C	0.585909	0.045287	2.123142
C	-2.602145	-0.762033	0.582330
C	-2.506823	0.630332	0.286686
C	-3.743350	-1.480152	0.133859
C	-4.728550	-0.785972	-0.576866
C	-4.624826	0.580583	-0.857115
C	-3.489193	1.317263	-0.420787
C	-0.925654	2.495573	0.755108
C	0.429400	2.774943	0.092430
C	0.713027	2.147984	-1.288220
C	2.528085	0.598314	-0.689990
C	2.784261	-0.787306	-0.551213
C	1.585908	-1.462445	-0.988604
C	0.653377	-0.455979	-1.358869
C	3.506644	1.551439	-0.374507
C	4.735495	1.092675	0.088010
C	4.991227	-0.278869	0.237136
C	4.019761	-1.226366	-0.083419
C	1.258197	-2.805735	-1.093230
C	-0.009430	-3.158601	-1.580928
C	-0.921986	-2.174714	-1.952027
C	-0.618229	-0.815493	-1.839039
C	-0.411807	2.313626	-2.295689
C	-1.005576	3.146510	2.137306
N	-1.298323	1.099112	0.803986
N	1.243388	0.787294	-1.170742
H	1.977514	-1.210985	3.225354
H	0.607595	-3.266429	3.004521
H	-1.572560	-3.237667	1.801501
H	1.192293	0.944099	2.240004
H	-3.855491	-2.543119	0.354408
H	-5.615451	-1.327763	-0.919282
H	-5.420310	1.090579	-1.403241
H	-3.430208	2.390601	-0.612552
H	-1.692569	2.971569	0.128799
H	0.500644	3.869121	-0.027377
H	1.245469	2.498234	0.775789
H	1.567535	2.717119	-1.689971
H	3.328973	2.622229	-0.481261
H	5.514039	1.815874	0.340227
H	5.964788	-0.605908	0.608726
H	4.223481	-2.292785	0.032686
H	1.972146	-3.575610	-0.793853
H	-0.284026	-4.212189	-1.661917
H	-1.916179	-2.455240	-2.303271
H	-1.368319	-0.081292	-2.119446
H	-0.151632	1.851039	-3.258492
H	-0.560644	3.389449	-2.471552
H	-1.369694	1.903293	-1.949913
H	-0.787093	4.223172	2.073331
H	-2.016385	3.021594	2.550689
H	-0.292966	2.687355	2.837644

1a (tg⁻) (-1227.856964 hartree)			
C	1.522216	-0.574270	3.141307
C	1.209254	-0.831125	1.665311
C	-0.156631	-0.263291	1.254103
C	-0.737347	-0.767637	-0.075813
C	0.066663	-0.442347	-1.333641
C	-3.221162	-1.158993	-0.103226
C	-4.405495	-0.386276	-0.209092
C	-3.985117	0.987357	-0.354032
C	-2.565829	0.983499	-0.330453
C	-4.680611	2.190777	-0.496963
C	-3.964495	3.377112	-0.611986
C	-2.562130	3.364352	-0.585107
C	-1.845242	2.179834	-0.446305
C	-3.280375	-2.548698	0.051272
C	-4.534823	-3.148793	0.096171
C	-5.712795	-2.393635	-0.008723
C	-5.653628	-1.012474	-0.161943
C	2.682113	0.944776	0.596100
C	3.773499	0.920441	-0.329118
C	4.050841	-0.448037	-0.631566
C	3.122636	-1.242150	0.109782
C	4.988338	-1.104211	-1.451529
C	4.969743	-2.513281	-1.499516
C	4.058062	-3.275736	-0.767805
C	3.101054	-2.631438	0.062351
C	2.169248	2.127101	1.123097
C	2.760780	3.351233	0.701762
C	3.815728	3.339437	-0.209353
C	4.339780	2.142856	-0.739429
N	2.301796	-0.376524	0.824038
N	-2.119739	-0.322259	-0.181878
H	0.747074	-1.033938	3.771319
H	1.562808	0.499853	3.368810
H	2.492981	-1.016711	3.406218
H	1.168305	-1.920523	1.523025
H	-0.122834	0.834360	1.252982
H	-0.866624	-0.547937	2.046611
H	-0.795264	-1.864665	-0.009345
H	-0.520674	-0.712768	-2.222961
H	0.999791	-1.021658	-1.361256
H	0.330780	0.621353	-1.406675
H	-5.773155	2.197101	-0.517332
H	-4.494410	4.325430	-0.724879
H	-2.013282	4.304562	-0.677857
H	-0.755179	2.204578	-0.432924
H	-2.379200	-3.159292	0.132756
H	-4.600291	-4.233030	0.214870
H	-6.681402	-2.896746	0.029966
H	-6.570344	-0.423541	-0.244636
H	5.714888	-0.539274	-2.037576
H	5.697837	-3.025753	-2.134399
H	4.076230	-4.364606	-0.828498
H	2.392565	-3.223755	0.643643
H	1.359526	2.138931	1.851691
H	2.385383	4.295137	1.099228
H	4.256244	4.290013	-0.522752
H	5.168204	2.166066	-1.449073

1a (g⁺g⁺) (-1227.845058 hartree)			
C	-0.372931	-0.910018	2.155506
C	0.646909	-1.491825	1.185808
C	-0.000119	-2.240575	-0.000288
C	-0.647056	-1.491566	-1.186313
C	0.372900	-0.909736	-2.155862
C	-1.783824	0.795104	-0.775639
C	-2.451945	3.498086	-0.543042
C	-0.853774	1.764168	-1.176648
C	-3.056675	1.176586	-0.247081
C	-3.380376	2.534741	-0.125752
C	-1.213301	3.126206	-1.063477
C	-2.941336	-1.106666	-0.340267
C	-5.484110	-1.625781	0.632649
C	-3.784265	-0.026572	0.030433
C	-3.357316	-2.433047	-0.245360
C	-4.649398	-2.681465	0.250321
C	-5.070312	-0.295593	0.526542
C	2.941343	-1.106644	0.340210
C	5.484533	-1.625410	-0.632075
C	3.357531	-2.433017	0.245044
C	3.784437	-0.026549	-0.030005
C	5.070573	-0.295367	-0.525730
C	4.649627	-2.681179	-0.250225
C	1.783749	0.794902	0.775449
C	2.451806	3.497985	0.543248
C	3.056654	1.176614	0.247458
C	0.853487	1.763935	1.176157
C	1.212988	3.125848	1.063183
C	3.380374	2.534736	0.126340
N	-1.727794	-0.593178	-0.801872
N	1.727714	-0.593390	0.801549
H	-0.961982	-1.739963	2.570917
H	-1.083125	-0.220632	1.684001
H	0.124060	-0.396142	2.990129
H	1.160751	-2.292986	1.746367
H	-0.752073	-2.903732	0.454031
H	0.751724	-2.903776	-0.454739
H	-1.160987	-2.292599	-1.746976
H	0.961840	-1.739710	-2.571381
H	1.083144	-0.220526	-1.684176
H	-0.123964	-0.395648	-2.990426
H	0.105727	1.507122	-1.612192
H	-4.350686	2.836587	0.274504
H	-0.503081	3.884962	-1.394245
H	-2.707962	4.557737	-0.464653
H	-2.721965	-3.265039	-0.553127
H	-5.002096	-3.711414	0.333481
H	-5.736482	0.519097	0.817725
H	-6.483091	-1.847052	1.016899
H	2.722115	-3.265147	0.552307
H	5.736763	0.519469	-0.816490
H	5.002382	-3.711105	-0.333530
H	6.483629	-1.846571	-1.016063
H	-0.106199	1.506837	1.611272
H	0.502536	3.884524	1.393654
H	4.350840	2.836599	-0.273539
H	2.707707	4.557667	0.464999

1a (g^+g^-) (-1227.851561 hartree)

C	-1.642251	-2.609892	-2.414310
C	-1.192648	-2.255269	-1.001856
C	0.314853	-1.982371	-0.845036
C	0.833118	-0.685274	-1.497486
C	1.432643	-0.866992	-2.892458
C	1.623891	1.306307	-0.207563
C	1.787660	3.839671	0.868201
C	0.563161	2.182118	-0.430115
C	2.787735	1.656875	0.555083
C	2.861485	2.948463	1.095276
C	0.659651	3.481146	0.127461
C	2.999597	-0.514783	-0.176439
C	5.467129	-1.037732	0.952142
C	3.646044	0.512637	0.571796
C	3.547369	-1.777146	-0.364868
C	4.822288	-2.033380	0.222890
C	4.907214	0.242473	1.143973
C	-2.274938	-1.272120	1.005284
C	-3.079892	-0.919699	3.644429
C	-1.918227	-2.237710	1.953043
C	-3.023384	-0.130494	1.375831
C	-3.427616	0.041548	2.703230
C	-2.331685	-2.045369	3.267636
C	-2.553914	-0.062370	-0.875106
C	-3.818499	2.387547	-1.366239
C	-3.198975	0.647181	0.174683
C	-2.545956	0.467848	-2.174166
C	-3.179558	1.687105	-2.398868
C	-3.829951	1.868504	-0.077622
N	1.783142	0.002655	-0.634873
N	-2.004042	-1.230341	-0.356977
H	-1.159056	-3.550654	-2.716500
H	-1.380247	-1.856974	-3.167195
H	-2.730969	-2.759340	-2.447761
H	-1.380563	-3.163182	-0.407196
H	0.863726	-2.856053	-1.231909
H	0.531115	-1.935749	0.232725
H	-0.015257	0.006070	-1.578615
H	0.704390	-1.346407	-3.560973
H	2.336424	-1.491890	-2.860150
H	1.705521	0.107775	-3.322025
H	-0.323679	1.893605	-0.997172
H	3.727084	3.264856	1.679616
H	-0.153538	4.192591	-0.020563
H	1.844885	4.846011	1.292173
H	3.036671	-2.554852	-0.930938
H	5.286063	-3.012914	0.098188
H	5.438471	1.002088	1.719229
H	6.443942	-1.254124	1.394012
H	-1.328709	-3.117944	1.690001
H	-4.006629	0.921047	2.995022
H	-2.061111	-2.787830	4.022259
H	-3.385040	-0.798784	4.686055
H	-2.063275	-0.040175	-3.006411
H	-3.175331	2.103106	-3.409325
H	-4.321031	2.408848	0.735294
H	-4.305632	3.341740	-1.577848

1a (g^-g^-) (-1227.855400 hartree)

C	-1.259591	-0.217759	1.479310
C	0.010200	0.017843	0.648569
C	1.283123	0.246760	1.477879
C	-3.428060	0.659106	0.571311
C	-2.711264	-1.281105	-0.322225
C	-3.576445	1.836058	1.313117
C	-4.394454	0.280086	-0.393496
C	-2.004186	-2.439255	-0.669890
C	-3.935572	-0.962962	-0.965664
C	-4.700590	2.621256	1.076569
C	-5.515982	1.085208	-0.612369
C	-2.537931	-3.264431	-1.654793
C	-4.451424	-1.809372	-1.951213
C	-5.664222	2.254437	0.124886
C	-3.749737	-2.959455	-2.291977
C	-1.183030	-1.398118	2.444937
C	2.726705	1.305769	-0.329173
C	3.421474	-0.675648	0.569647
C	2.016090	2.458186	-0.651476
C	3.958242	0.950196	-0.967533
C	3.514371	-1.843259	1.317653
C	4.391795	-0.290337	-0.405764
C	2.555941	3.309717	-1.653821
C	4.476944	1.807195	-1.955063
C	4.639996	-2.681830	1.088907
C	5.499716	-1.132335	-0.617511
C	3.757525	2.977281	-2.277903
C	5.597438	-2.317655	0.141720
C	1.232461	1.437181	2.433406
N	-2.420396	-0.293543	0.609513
N	2.438442	0.308578	0.598381
H	-0.165424	0.879951	-0.012594
H	0.190177	-0.844270	-0.011897
H	-1.425233	0.682333	2.088349
H	1.447276	-0.650154	2.089350
H	-2.842333	2.146284	2.059296
H	-1.057678	-2.703290	-0.198284
H	-4.832281	3.544180	1.646568
H	-6.265019	0.797899	-1.354297
H	-1.994839	-4.169676	-1.936675
H	-5.394154	-1.566104	-2.447362
H	-6.534748	2.894084	-0.035798
H	-4.139561	-3.629550	-3.061228
H	-0.429778	-1.216944	3.226228
H	-0.923045	-2.338972	1.939910
H	-2.152061	-1.539227	2.944616
H	1.073422	2.714671	-0.169876
H	2.762746	-2.124987	2.056794
H	2.023639	4.220403	-1.931193
H	5.413580	1.574177	-2.463895
H	4.749862	-3.607051	1.656007
H	6.267072	-0.878705	-1.350487
H	4.157895	3.644171	-3.046496
H	6.456944	-2.974079	-0.020292
H	0.983345	2.377437	1.923961
H	2.205300	1.566048	2.928755
H	0.481385	1.259579	3.217039

1b (tt #1) (-1855.667716 hartree)

C	-2.428759	-0.418867	1.004830
C	0.233722	-0.680051	-1.509284
C	-2.054715	-1.790086	0.932658
C	-0.216450	-0.630566	1.584295
C	4.740269	-0.892828	-0.386916
C	2.452195	-0.390850	-1.018692
C	1.252110	0.298446	-1.435960
C	1.059201	-0.239841	1.981449
C	0.975981	1.622364	-1.777536
C	1.267757	1.118638	2.287733
C	0.011238	-3.159983	-1.212961
C	3.759888	0.032808	-0.772788
C	2.109030	-1.760551	-0.886394
C	-6.119603	-0.591243	-0.028924
C	-4.689089	-1.028343	0.307640
C	-3.751801	-0.045944	0.698248
C	-1.272300	0.310883	1.433259
C	0.070962	-3.954107	0.098693
C	-1.037118	1.661138	1.745073
C	-4.288759	-2.368799	0.246078
C	-0.319226	1.986402	-2.191934
C	0.235058	2.071185	2.192220
C	-0.666095	3.406706	-2.648366
C	3.073973	-2.701776	-0.519641
C	-1.319412	0.995609	-2.216068
C	-1.073566	-0.332546	-1.867692
C	6.186140	-0.480402	-0.092054
C	-2.964787	-2.773083	0.562661
C	4.371260	-2.248926	-0.271877
C	0.075437	-3.106649	1.380663
C	0.521265	4.363688	-2.522301
C	-1.092691	3.363091	-4.125923
C	-1.823646	3.957314	-1.800650
C	0.519907	3.514910	2.623163
C	-0.323851	-3.921796	2.608834
C	0.396503	-4.008200	-2.422025
C	-6.085579	0.406810	-1.198160
C	-6.747341	0.085799	1.200490
C	-0.695334	4.429137	2.452450
C	6.404003	1.024263	-0.267303
C	6.533987	-0.850784	1.358992
C	7.133519	-1.219198	-1.051301
C	0.914595	3.524072	4.110249
C	1.675727	4.090389	1.789545
C	-7.013295	-1.765994	-0.433115
N	0.760467	-1.913455	-1.169691
N	-0.703609	-1.889115	1.266266
H	1.878883	-0.952439	2.088435
H	1.774936	2.362066	-1.731185
H	2.259896	1.422729	2.625721
H	-1.035929	-2.850872	-1.335596
H	4.001544	1.089970	-0.880921
H	-4.041351	1.004115	0.767779
H	-0.786077	-4.645176	0.090825
H	0.961731	-4.601598	0.129744
H	-1.856541	2.374421	1.661271
H	-5.000904	-3.141537	-0.040858
H	2.842966	-3.763056	-0.428979
H	-2.336055	1.261457	-2.511202
H	-1.883349	-1.062550	-1.898191
H	-2.701959	-3.830487	0.531037
H	5.123318	-2.986471	0.016568
H	1.109077	-2.763491	1.527207
H	0.870180	4.447989	-1.481974
H	0.224416	5.370322	-2.854310
H	1.371389	4.047923	-3.146262
H	-0.280897	2.972443	-4.758916
H	-1.346868	4.375004	-4.480902
H	-1.974018	2.722464	-4.277281
H	-2.729699	3.340226	-1.889320
H	-2.081150	4.979088	-2.123113
H	-1.547852	3.989323	-0.735947
H	-0.200496	-3.321446	3.521537
H	0.310220	-4.817279	2.695529
H	-1.374004	-4.243405	2.549602
H	0.249184	-3.442248	-3.353136
H	-0.233717	-4.909058	-2.465183
H	1.448317	-4.327407	-2.377621
H	-5.484953	1.296549	-0.957760
H	-7.104736	0.744762	-1.448539
H	-5.650234	-0.058681	-2.096508
H	-6.786065	-0.610627	2.052343
H	-7.776254	0.413400	0.977779
H	-6.173801	0.968632	1.518690
H	-1.023649	4.482348	1.403522
H	-0.441384	5.451526	2.772955
H	-1.548236	4.092115	3.060680
H	6.199493	1.350795	-1.298441
H	7.451558	1.276822	-0.042376
H	5.768416	1.612223	0.412483
H	5.868757	-0.333128	2.067368
H	7.571610	-0.560903	1.590748
H	6.441301	-1.931853	1.540916
H	7.058841	-2.311466	-0.942972
H	8.179355	-0.934695	-0.852064
H	6.905471	-0.970063	-2.099160
H	0.103983	3.113614	4.731906
H	1.118571	4.553533	4.448149
H	1.815821	2.921860	4.296643
H	2.596236	3.499037	1.901554
H	1.898713	5.124064	2.101162
H	1.416661	4.102840	0.719664
H	-6.630175	-2.281543	-1.327198
H	-8.025914	-1.401631	-0.666725
H	-7.104437	-2.505993	0.376654

1b (tt-#2) (-1855.649113 hartree)

C	-1.726315	0.137985	0.987448
C	-5.614849	-2.091541	-0.355093
C	-2.943904	-1.701985	0.427970
C	-5.141767	-0.777010	-0.093710
C	1.733962	0.119184	-1.088746
C	3.757220	-0.636250	-0.326949
C	3.031635	0.547950	-0.718948
C	-3.807812	-0.577669	0.302966
C	-7.072499	-2.272198	-0.784428
C	5.522130	-2.131537	0.409678
C	-3.048970	0.581389	0.663718
C	2.444649	2.842416	-1.260002
C	-0.061670	-2.868970	-0.093707
C	-0.512150	-2.044283	1.115727
C	3.372172	1.900398	-0.802620
C	0.467155	-2.044796	-1.271920
C	-0.729599	1.020361	1.401308
C	2.845816	4.320929	-1.349571
C	6.945357	-2.418692	0.902047
C	5.066814	-0.845133	0.120735
C	-4.744022	-3.173704	-0.208972
C	-3.351196	1.944407	0.778568
C	2.868503	-1.728100	-0.484436
C	3.313295	-3.027643	-0.213246
C	-3.387970	-2.995474	0.189855
C	-2.358186	2.857747	1.213537
C	-1.062786	2.386002	1.512073
C	4.620875	-3.202449	0.226799
C	1.161284	2.383949	-1.617158
C	-2.664734	4.346466	1.384894
C	0.788523	1.042432	-1.537530
C	1.741257	5.184927	-1.963188
C	3.144730	4.849012	0.062673
C	-0.735639	-2.891891	2.367538
C	0.705380	-2.902546	-2.512470
C	4.101094	4.464906	-2.224411
C	-7.317486	-1.491163	-2.086589
C	-7.996763	-1.729326	0.318695
C	-4.100268	4.701851	0.989816
C	-2.462282	4.730604	2.861374
C	-1.707958	5.171547	0.507913
C	7.772274	-1.139075	1.045260
C	-7.434301	-3.738281	-1.031728
C	7.657433	-3.341380	-0.099954
C	6.882795	-3.106044	2.275354
N	1.637817	-1.256044	-0.923174
N	-1.685664	-1.230545	0.826645
H	-5.808116	0.081344	-0.192905
H	0.711808	-3.566316	0.258889
H	-0.892036	-3.486961	-0.462652
H	0.293670	-1.329655	1.334796
H	-0.313544	-1.311435	-1.524069
H	0.280675	0.685877	1.639758
H	5.725081	0.017245	0.234023
H	-5.089544	-4.188991	-0.396747
H	-4.357562	2.289363	0.545580
H	2.674571	-3.900601	-0.349265
H	-2.748758	-3.868592	0.313678
H	-0.288869	3.081087	1.838644
H	4.954413	-4.222580	0.432523
H	0.413116	3.092658	-1.972352
H	-0.220622	0.745884	-1.828661
H	0.820816	5.167162	-1.360393
H	1.489084	4.859574	-2.984323
H	2.077258	6.231844	-2.021411
H	3.442026	5.909903	0.027020
H	3.960487	4.287499	0.542663
H	2.256696	4.764522	0.708575
H	-1.030699	-2.254051	3.212985
H	0.198750	-3.406822	2.634456
H	-1.520849	-3.646182	2.218607
H	1.055278	-2.277315	-3.346398
H	-0.228228	-3.396579	-2.820199
H	1.461066	-3.681567	-2.334407
H	3.916040	4.088747	-3.242464
H	4.956905	3.909054	-1.813474
H	4.398458	5.523481	-2.299554
H	-7.114223	-0.417019	-1.966001
H	-8.366085	-1.601834	-2.408000
H	-6.670266	-1.865044	-2.894995
H	-7.842494	-2.275307	1.262201
H	-9.052804	-1.844534	0.024354
H	-7.817485	-0.662536	0.517269
H	-4.305014	4.464493	-0.065312
H	-4.265324	5.781757	1.125603
H	-4.839101	4.173743	1.611431
H	-3.136297	4.153201	3.512682
H	-2.677046	5.801556	3.009227
H	-1.431401	4.542839	3.194991
H	-0.655455	5.004248	0.778409
H	-1.918621	6.247160	0.623128
H	-1.827771	4.912217	-0.555464
H	8.783459	-1.386626	1.403497
H	7.325558	-0.441950	1.770856
H	7.881850	-0.612377	0.084792
H	-6.820805	-4.180176	-1.831717
H	-8.488670	-3.814909	-1.339532
H	-7.307480	-4.348284	-0.124312
H	7.719741	-2.869727	-1.092847
H	7.132312	-4.301023	-0.218614
H	8.682626	-3.562237	0.239174
H	7.897959	-3.324813	2.644616
H	6.332145	-4.057784	2.232974
H	6.381568	-2.462319	3.014768
H	4.379314	2.211871	-0.513806

1b (tg$^+$) (-1857.576788 hartree)

C	-0.616534	3.594049	-2.231471
C	0.618925	3.275211	-1.412739
C	0.602251	3.868807	0.006727
C	-0.661193	3.679173	0.848199
C	-0.518485	4.377158	2.199502
C	2.327825	1.516990	-1.201695
C	4.828752	0.328040	-0.835104
C	3.430599	2.331447	-0.956178
C	2.473057	0.121995	-1.282556
C	3.711782	-0.464759	-1.098252
C	4.656879	1.721231	-0.778099
C	0.283074	0.708268	-1.648515
C	-0.661004	-1.900547	-2.089758
C	1.161549	-0.399174	-1.583351
C	-1.071124	0.500305	-1.909234
C	-1.509735	-0.798937	-2.132799
C	0.698144	-1.674401	-1.800755
C	-2.336090	1.851914	0.671266
C	-4.711775	0.448556	0.204878
C	-3.391985	2.520912	0.078710
C	-2.427384	0.484882	1.038008
C	-3.635079	-0.211728	0.800062
C	-4.583774	1.796958	-0.155185
C	-0.354907	1.270438	1.611690
C	0.576264	-1.117006	2.746245
C	-1.191244	0.124333	1.627601
C	0.918785	1.237348	2.140794
C	1.376675	0.027857	2.708569
C	-0.710133	-1.068277	2.204399
C	6.213732	-0.263804	-0.603241
C	6.699866	0.125773	0.798785
C	7.187538	0.290053	-1.650987
C	6.217201	-1.786800	-0.703617
C	-1.148080	-3.323278	-2.319395
C	-0.878589	-4.154970	-1.058246
C	-2.642077	-3.382285	-2.623481
C	-0.388181	-3.934814	-3.503778
C	-6.043798	-0.257163	-0.051074
C	-6.035853	-1.713689	0.406139
C	-7.162110	0.466304	0.709679
C	-6.362489	-0.234322	-1.550871
C	1.131174	-2.388579	3.389276
C	2.385560	-2.843230	2.633028
C	1.501314	-2.108774	4.850858
C	0.129880	-3.540933	3.372790
N	-1.064939	2.303795	0.997771
N	1.008284	1.862725	-1.420711

H	-0.673750	4.678686	-2.346992
H	-1.540078	3.265194	-1.757388
H	-0.553261	3.155172	-3.228425
H	1.458038	3.765256	-1.918229
H	0.754911	4.948197	-0.100993
H	1.469047	3.492026	0.552281
H	-1.482813	4.165291	0.319696
H	-0.277520	5.434177	2.063512
H	0.264916	3.916351	2.802932
H	-1.456572	4.304929	2.751654
H	3.350667	3.409200	-0.895313
H	3.804050	-1.541205	-1.154501
H	5.515137	2.352089	-0.580794
H	-1.792988	1.300853	-1.925369
H	-2.565904	-0.937001	-2.313307
H	1.386700	-2.509147	-1.739488
H	-3.350755	3.575832	-0.167429
H	-3.715266	-1.246569	1.103416
H	-5.419526	2.318251	-0.604905
H	1.564925	2.105508	2.155811
H	2.374181	0.007667	3.129832
H	-1.358280	-1.933788	2.235671
H	7.693752	-0.292518	0.979511
H	6.023669	-0.256948	1.566743
H	6.766928	1.208701	0.919863
H	8.186293	-0.124096	-1.488699
H	7.266293	1.377732	-1.599753
H	6.868540	0.023297	-2.661475
H	7.228717	-2.159638	-0.527644
H	5.908155	-2.130496	-1.694105
H	5.563167	-2.245306	0.042197
H	-1.402157	-3.734658	-0.197157
H	0.185181	-4.195632	-0.815037
H	-1.226406	-5.180702	-1.207889
H	-3.238614	-2.986061	-1.798562
H	-2.939842	-4.421713	-2.779324
H	-2.894920	-2.826376	-3.529845
H	0.689350	-3.961686	-3.328437
H	-0.566571	-3.365757	-4.419437
H	-0.723398	-4.962069	-3.670281
H	-5.278086	-2.299425	-0.120509
H	-7.009172	-2.167762	0.201142
H	-5.846844	-1.798249	1.478795
H	-7.258152	1.505715	0.391356
H	-6.960311	0.464372	1.783583
H	-8.122466	-0.031191	0.539891
H	-5.590502	-0.761253	-2.118585
H	-6.420141	0.785893	-1.934256
H	-7.321396	-0.725261	-1.745733
H	3.157995	-2.071730	2.640504
H	2.144702	-3.065191	1.589707
H	2.801789	-3.747694	3.088551
H	2.255299	-1.323828	4.931711
H	1.899137	-3.012175	5.324375
H	0.623036	-1.785096	5.414602
H	-0.151351	-3.817686	2.353632
H	-0.780790	-3.290826	3.922094
H	0.574606	-4.420492	3.846457

1b (tg$^-$) (-1857.582175 hartree)

C	1.324909	-0.854147	3.618465
C	1.136081	-1.127078	2.127156
C	-0.168788	-0.524429	1.596981
C	-0.681008	-1.085843	0.265675
C	0.206498	-0.828529	-0.946844
C	-3.158346	-1.437376	0.084817
C	-4.312797	-0.656506	-0.130545
C	-3.860910	0.699808	-0.299923
C	-2.458089	0.676088	-0.180525
C	-4.528778	1.898140	-0.536531
C	-3.822518	3.085568	-0.654846
C	-2.424183	3.031138	-0.527632
C	-1.730684	1.858758	-0.295254
C	-3.268395	-2.806463	0.285659
C	-4.533455	-3.374109	0.266802
C	-5.698623	-2.626217	0.054242
C	-5.564325	-1.255466	-0.144462
C	2.756517	0.589826	1.190680
C	3.932023	0.525888	0.393930
C	4.195809	-0.848810	0.131007
C	3.174733	-1.600254	0.768767
C	5.191582	-1.532256	-0.571005
C	5.168074	-2.946226	-0.631750
C	4.149545	-3.648837	0.008489
C	3.128502	-2.979481	0.722604
C	2.243952	1.790122	1.645069
C	2.920253	2.980073	1.284393
C	4.067083	2.955116	0.496804
C	4.578313	1.714551	0.047009
C	-4.500081	4.428398	-0.915466
C	-4.185456	5.393655	0.233599
C	-3.976596	5.020368	-2.229496
C	-6.016750	4.297223	-1.024199
C	-7.088012	-3.261777	0.033542
C	-7.041246	-4.769466	0.267978
C	-7.954077	-2.635614	1.132928
C	-7.744725	-3.012055	-1.329246
C	4.800831	4.229681	0.102044
C	4.139586	5.488160	0.657054
C	6.239348	4.172284	0.634184
C	4.833058	4.342717	-1.428303
C	6.271065	-3.657256	-1.404219
C	7.631264	-3.303290	-0.787307
C	6.244596	-3.194153	-2.867391
C	6.122411	-5.176057	-1.384719
N	2.311812	-0.710453	1.385078
N	-2.041254	-0.624172	0.048206

H	0.494638	-1.289764	4.177137
H	1.361896	0.214861	3.830099
H	2.254067	-1.304424	3.971056
H	1.081316	-2.209288	1.994167
H	-0.080213	0.559780	1.530843
H	-0.935396	-0.728211	2.349985
H	-0.764107	-2.169740	0.380184
H	-0.313717	-1.145844	-1.852219
H	1.133113	-1.399166	-0.875889
H	0.468760	0.224044	-1.055038
H	-5.607953	1.887858	-0.626038
H	-1.848900	3.945877	-0.616406
H	-0.652001	1.885437	-0.210901
H	-2.402819	-3.436856	0.451486
H	-4.606048	-4.442485	0.423157
H	-6.441937	-0.640364	-0.313144
H	5.978804	-0.977770	-1.065723
H	4.123247	-4.727849	-0.029742
H	2.354999	-3.553074	1.217592
H	1.367392	1.847745	2.275031
H	2.522426	3.918915	1.640906
H	5.471694	1.680937	-0.563585
H	-3.111673	5.561895	0.338147
H	-4.659404	6.363322	0.055389
H	-4.556688	5.001813	1.183806
H	-4.449517	5.987127	-2.425393
H	-2.896459	5.177410	-2.200682
H	-4.195297	4.356781	-3.069695
H	-6.309317	3.643459	-1.849809
H	-6.457587	3.907292	-0.103152
H	-6.456285	5.280286	-1.210069
H	-6.462843	-5.283329	-0.503983
H	-6.611472	-5.016573	1.242073
H	-8.056174	-5.173865	0.243371
H	-8.062016	-1.557606	0.996858
H	-8.955634	-3.075566	1.124995
H	-7.514525	-2.805285	2.118953
H	-7.846881	-1.945320	-1.538738
H	-7.152604	-3.454901	-2.133856
H	-8.744435	-3.455414	-1.356689
H	3.117135	5.603738	0.289245
H	4.114380	5.483918	1.749436
H	4.707105	6.367227	0.341631
H	6.785690	3.315236	0.236819
H	6.780291	5.079473	0.349072
H	6.246710	4.097820	1.724274
H	5.367332	5.249968	-1.725519
H	5.335283	3.489938	-1.888146
H	3.820112	4.393738	-1.834946
H	8.433416	-3.801800	-1.339506
H	7.821927	-2.229105	-0.812551
H	7.680505	-3.627696	0.254921
H	7.039314	-3.691213	-3.431517
H	5.288081	-3.440173	-3.334831
H	6.392402	-2.116499	-2.955153
H	6.165029	-5.573879	-0.367908
H	5.182937	-5.496668	-1.841419
H	6.938861	-5.628990	-1.952670

1b (g^+g^+) (-1855.648276 hartree)

C	2.203916	0.497039	0.787700
C	3.568739	0.601462	0.350675
C	4.116215	1.868547	0.101360
C	3.335432	3.032400	0.295721
C	2.005661	2.899876	0.743006
C	1.426881	1.636813	0.997168
C	3.063457	-1.609119	0.651894
C	4.102499	-0.721357	0.269038
C	3.252877	-2.979679	0.733957
C	4.539632	-3.493938	0.400270
C	5.582430	-2.652422	0.003445
C	5.362866	-1.249044	-0.064864
C	6.967456	-3.192465	-0.359771
C	7.054262	-4.715495	-0.240033
C	7.296927	-2.805832	-1.811662
C	8.015023	-2.576704	0.583286
C	3.901071	4.430030	0.041265
C	5.351125	4.397338	-0.447636
C	3.048544	5.134014	-1.028594
C	3.851843	5.239633	1.348784
C	0.663127	-1.524791	1.259073
C	0.048382	-2.270263	0.051824
C	-0.588586	-1.546163	-1.150396
C	0.389740	-0.708751	-1.961854
C	-0.334791	-0.702221	2.058318
C	-3.000316	-1.632394	-0.595411
C	-4.074974	-0.769294	-0.278607
C	-3.548634	0.572074	-0.349988
C	-2.181651	0.459342	-0.708723
C	-4.139749	1.823668	-0.145999
C	-3.395471	2.992699	-0.294445
C	-2.039795	2.855549	-0.660139
C	-1.422968	1.627415	-0.869806
C	-3.203565	-3.013275	-0.641893
C	-4.476683	-3.506199	-0.361271
C	-5.562997	-2.672728	-0.032215
C	-5.337832	-1.292480	0.000542
C	-3.986700	4.390821	-0.084357
C	-5.464435	4.335242	0.309116
C	-3.217280	5.108202	1.036233
C	-3.865057	5.198333	-1.386436
C	-6.960641	-3.223015	0.280966
C	-7.014385	-4.750265	0.194341
C	-7.364714	-2.809094	1.704871
C	-7.972310	-2.650245	-0.724346
N	1.917405	-0.851897	0.930775
N	-1.854955	-0.887929	-0.841851
H	5.147324	1.941716	-0.241479
H	1.394155	3.786842	0.913930
H	0.403066	1.589170	1.354255
H	2.465756	-3.662790	1.056249
H	4.692173	-4.569942	0.464167
H	6.167505	-0.577107	-0.367376
H	8.066266	-5.053336	-0.512101
H	6.854916	-5.055554	0.787655
H	6.343039	-5.218611	-0.912814
H	8.295981	-3.179569	-2.089755
H	7.292691	-1.715809	-1.958021
H	6.562974	-3.239551	-2.508381
H	7.802531	-2.841882	1.630471
H	9.021692	-2.949289	0.332429
H	8.035712	-1.479245	0.514575
H	5.708425	5.424601	-0.617946
H	6.021283	3.931449	0.290666
H	5.450614	3.849444	-1.397019
H	3.440627	6.145620	-1.222773
H	1.998804	5.233889	-0.716412
H	3.065708	4.573215	-1.975884
H	4.453272	4.754848	2.133100
H	4.254898	6.252282	1.184713
H	2.825265	5.342492	1.729279
H	0.988337	-2.340978	1.927135
H	0.828629	-2.926645	-0.364259
H	-0.711319	-2.939385	0.486271
H	-0.879872	-2.376962	-1.817633
H	-0.109523	-0.258946	-2.831793
H	1.191111	-1.365402	-2.330309
H	0.870340	0.087622	-1.381161
H	-1.106489	-1.384810	2.442291
H	0.150206	-0.218739	2.917908
H	-0.850389	0.059292	1.462058
H	-5.194459	1.864185	0.129703
H	-1.426947	3.750824	-0.793027
H	-0.374000	1.609519	-1.152188
H	-2.402242	-3.709484	-0.896306
H	-4.619695	-4.586416	-0.403851
H	-6.153953	-0.606210	0.241269
H	-5.851156	5.356015	0.452432
H	-6.077737	3.855902	-0.469406
H	-5.616825	3.788401	1.252448
H	-3.624872	6.119180	1.199043
H	-2.148807	5.215177	0.795840
H	-3.293215	4.552595	1.983831
H	-2.817306	5.308774	-1.703743
H	-4.412281	4.707489	-2.206063
H	-4.281878	6.210005	-1.254031
H	-8.031968	-5.100210	0.427005
H	-6.763187	-5.113463	-0.814172
H	-6.329272	-5.226826	0.912481
H	-8.369537	-3.190976	1.948271
H	-7.386608	-1.715286	1.822447
H	-6.657507	-3.210595	2.447156
H	-7.707371	-2.934382	-1.754514
H	-8.984775	-3.031364	-0.513645
H	-8.014685	-1.551593	-0.681935

1b (g^+g^-) (-1855.656139 hartree)

C	0.969402	-4.051083	-2.794251
C	0.772220	-2.545718	-2.654672
C	-0.598621	-2.132752	-2.052687
C	-0.665923	-2.143045	-0.520661
C	-0.919123	-3.487785	0.151247
C	2.022494	-0.459215	-2.270795
C	2.689556	2.275087	-2.328139
C	1.506607	0.329115	-3.284696
C	2.859222	0.082246	-1.252967
C	3.192146	1.468169	-1.300237
C	1.851553	1.708336	-3.306703
C	2.563586	-2.166967	-0.857900
C	4.143507	-2.262231	1.490615
C	3.189361	-0.975830	-0.367702
C	2.735830	-3.376496	-0.202318
C	3.525777	-3.411368	0.982816
C	3.977717	-1.043334	0.813557
C	-1.269574	0.031788	0.558288
C	-1.049142	2.497917	1.865860
C	0.019041	0.543440	0.825858
C	-2.442012	0.760557	0.918974
C	-2.330311	1.972563	1.569944
C	0.094689	1.763063	1.465901
C	-3.015288	-1.189231	-0.143024
C	-5.800479	-0.822691	-0.062696
C	-3.568543	-0.037616	0.459106
C	-3.840573	-2.154139	-0.720343
C	-5.217562	-1.950809	-0.667158
C	-4.943153	0.141925	0.502811
C	3.058108	3.763176	-2.440749
C	1.785115	4.619106	-2.522658
C	3.896607	3.980339	-3.710944
C	3.878472	4.254736	-1.244921
C	5.012605	-2.299141	2.756941
C	4.434840	-1.338548	3.808172
C	5.080284	-3.694748	3.381501
C	6.445352	-1.865975	2.408301
C	-0.864498	3.818826	2.603353
C	-2.195486	4.510846	2.906370
C	-0.145216	3.534359	3.935010
C	-0.005157	4.762684	1.744451
C	-7.323386	-0.674448	-0.045158
C	-7.841351	-0.636872	-1.492640
C	-7.938575	-1.875033	0.692788
C	-7.769696	0.607338	0.662016
N	-1.626801	-1.129695	-0.052776
N	1.857984	-1.830869	-2.028784

H	0.248044	-4.441993	-3.527847
H	0.820711	-4.605692	-1.860172
H	1.982022	-4.268679	-3.161276
H	0.738277	-2.164732	-3.688509
H	-1.396965	-2.752826	-2.495082
H	-0.778799	-1.096014	-2.375646
H	0.306331	-1.785447	-0.157661
H	0.001947	-4.079446	0.089940
H	-1.733949	-4.061309	-0.312708
H	-1.144386	-3.355228	1.219626
H	0.872547	-0.081784	-4.074352
H	3.860345	1.872838	-0.539831
H	1.463231	2.324597	-4.119755
H	2.328291	-4.311070	-0.583036
H	3.652706	-4.373731	1.477261
H	4.465371	-0.134564	1.173831
H	0.934004	0.034419	0.513202
H	-3.228641	2.522035	1.849771
H	1.088955	2.169552	1.650886
H	-3.436294	-3.038895	-1.209548
H	-5.860858	-2.706565	-1.121690
H	-5.351528	1.036252	0.973587
H	2.039480	5.690720	-2.585180
H	1.152546	4.466844	-1.633484
H	1.179874	4.366214	-3.405048
H	4.177473	5.041820	-3.822124
H	3.344165	3.676706	-4.612475
H	4.819775	3.381849	-3.671498
H	4.102258	5.327883	-1.356754
H	4.836947	3.721272	-1.161112
H	3.334682	4.120734	-0.296239
H	3.410355	-1.634975	4.084039
H	4.393132	-0.305636	3.432361
H	5.051107	-1.342236	4.723311
H	4.082681	-4.058412	3.672642
H	5.707455	-3.671602	4.287185
H	5.520545	-4.427909	2.688611
H	6.470749	-0.849520	1.989045
H	6.883431	-2.543044	1.658618
H	7.088812	-1.881592	3.304480
H	-2.751923	4.755504	1.988223
H	-2.004444	5.456404	3.434725
H	-2.842213	3.900913	3.555898
H	0.838696	3.068309	3.780859
H	-0.740786	2.864623	4.574262
H	0.011657	4.477464	4.480997
H	-0.479858	4.963569	0.772124
H	0.996591	4.354328	1.549522
H	0.124368	5.723020	2.266616
H	-7.594491	-1.556549	-2.043481
H	-7.412265	0.212533	-2.046032
H	-8.937243	-0.529287	-1.499633
H	-7.694972	-2.830173	0.204393
H	-9.035732	-1.783812	0.714079
H	-7.581121	-1.927660	1.732716
H	-7.387244	1.509998	0.161015
H	-7.448684	0.630846	1.714949
H	-8.867982	0.669857	0.653149

1b (g^-g^-) (-1855.657993 hartree)

C	-1.232266	0.305770	-2.230495
C	0.016547	-0.025990	-1.400951
C	1.269975	-0.347083	-2.228363
C	-3.464750	-0.398382	-1.326357
C	-2.597571	1.472958	-0.425056
C	-3.719987	-1.555049	-2.063194
C	-4.399426	0.050320	-0.362185
C	-1.815896	2.572006	-0.061804
C	-3.842614	1.252433	0.216706
C	-4.907518	-2.244073	-1.825318
C	-5.579590	-0.660735	-0.147327
C	-2.291294	3.429253	0.928306
C	-4.291108	2.129848	1.203335
C	-5.857593	-1.823079	-0.875124
C	-3.524367	3.237596	1.579497
C	-1.065397	1.479927	-3.192402
C	2.626098	-1.513106	-0.419128
C	3.475349	0.399055	-1.318375
C	1.839183	-2.608725	-0.087013
C	3.878638	-1.257945	0.221835
C	3.676325	1.553189	-2.061207
C	4.410893	-0.056523	-0.342314
C	2.313349	-3.493952	0.917365
C	4.326072	-2.148935	1.209841
C	4.865864	2.294443	-1.829150
C	5.579915	0.692607	-0.131042
C	3.537248	-3.278757	1.562812
C	5.810561	1.879078	-0.881180
C	1.130183	-1.528667	-3.185579
C	7.095882	2.666274	-0.621546
C	7.209722	3.914090	-1.500031
C	8.308720	1.764516	-0.909477
C	7.127888	3.111225	0.850880
C	-7.162880	-2.587262	-0.618061
C	-8.359119	-1.664799	-0.901515
C	-7.206026	-3.044524	0.848679
C	-7.293785	-3.825462	-1.508320
C	-4.040069	4.186506	2.669002
C	-3.065662	5.334086	2.945180
C	-5.381197	4.791901	2.225076
C	-4.240341	3.402432	3.975667
C	4.059689	-4.223724	2.645552
C	5.413044	-4.804194	2.199964
C	4.246658	-3.438339	3.955250
C	3.106415	-5.389014	2.919140
N	-2.383000	0.469589	-1.361043
N	2.416489	-0.500053	-1.348143
H	-0.223098	-0.874372	-0.742082
H	0.258733	0.818884	-0.737997
H	-1.466823	-0.578140	-2.841127
H	1.503363	0.536647	-2.837029
H	-3.020953	-1.931069	-2.813072
H	-0.849341	2.775150	-0.523340
H	-5.092612	-3.146411	-2.408944
H	-6.288087	-0.296802	0.601335
H	-1.665109	4.279942	1.198862
H	-5.254525	1.937867	1.682842
H	-0.322399	1.248339	-3.970223
H	-0.741618	2.397417	-2.681034
H	-2.019158	1.692357	-3.696251
H	0.878876	-2.803090	-0.563347
H	2.958658	1.902028	-2.805704
H	5.280876	-1.967056	1.705685
H	5.028490	3.200595	-2.409848
H	6.307698	0.359297	0.610410
H	0.817111	-2.448853	-2.674409
H	2.089317	-1.726469	-3.684971
H	0.389761	-1.298164	-3.965594
H	7.218929	3.659678	-2.570870
H	6.381736	4.616657	-1.320159
H	8.149131	4.442460	-1.275310
H	9.245642	2.313766	-0.720917
H	8.312742	0.865212	-0.276543
H	8.310504	1.434336	-1.959721
H	8.052052	3.674844	1.059002
H	6.270347	3.761785	1.082017
H	7.093207	2.255283	1.540471
H	-9.307865	-2.194963	-0.718483
H	-8.348344	-0.771007	-0.259877
H	-8.354195	-1.324113	-1.948483
H	-8.139477	-3.592820	1.055442
H	-6.360034	-3.710596	1.078494
H	-7.158698	-2.193432	1.544430
H	-7.297514	-3.563857	-2.577773
H	-6.479854	-4.545718	-1.332599
H	-8.242293	-4.340453	-1.291268
H	-2.088730	4.967758	3.296522
H	-2.899473	5.955780	2.051852
H	-3.474799	5.987945	3.730731
H	-6.142508	4.016917	2.050179
H	-5.771531	5.474460	2.997376
H	-5.265450	5.362995	1.290890
H	-3.292667	2.958527	4.317652
H	-4.614068	4.067026	4.771489
H	-4.967376	2.585248	3.855799
H	1.694921	-4.351063	1.177447
H	6.161616	-4.017368	2.026760
H	5.809725	-5.482878	2.972662
H	5.305444	-5.375215	1.264871
H	3.290301	-3.013379	4.296924
H	4.629218	-4.103103	4.746943
H	4.958835	-2.608352	3.839691
H	2.123677	-5.038427	3.269523
H	2.951393	-6.008368	2.022543
H	3.527284	-6.037570	3.702996

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